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**A Multi-Dimensional Framework for Evaluating
the Justifiability of AI Use Cases**

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Summary

The rapid proliferation of artificial intelligence (AI) systems across all sectors of society has created an urgent need for principled methods of evaluating whether individual deployments are genuinely justifiable. Existing responsible AI frameworks tend to address single dimensions of risk — most commonly fairness, transparency, or environmental impact — in isolation, leaving practitioners without a unified means of comparing and ranking AI deployments across multiple competing concerns.

This project presents the AI Misuse Evaluator: a reproducible, modular Python framework that integrates five normalised dimensions — environmental impact, ethical risk, creative displacement, purpose justification, and transparency/explainability — into a single Justifiability Score on a 0-100 scale. The framework applies a configurable weighted sum model with a non-compensatory guardrail for harmful use cases, and includes a full robustness validation suite comprising sensitivity analysis, Monte Carlo weight sampling, and Pareto frontier identification.

Evaluating 40 real-world AI use cases drawn from healthcare, infrastructure, creative industries, and adversarial contexts, the framework correctly places essential, beneficial, low-benefit, and harmful use cases into distinct score bands. Mean scores by category are: essential 81.57, beneficial 67.15, low-benefit 42.94, and harmful 9.14. Face validity testing against 15 hand-labelled ground truth cases yielded 87% accuracy (13/15). Sensitivity analysis across five weight configurations produced Spearman rank stability scores of 0.986-1.000, confirming that the ranking of use cases is robust to reasonable changes in analyst priorities. Monte Carlo robustness testing across 200 randomly sampled weight vectors found that `Earthquake_Early_Warning` and `Accessibility_Captioning` jointly dominate the top rank (52% and 48% of runs respectively). An inter-rater reliability study with two non-expert raters across 12 use cases yielded Cohen's weighted kappa of 0.955 (ethical risk) and 0.720 (transparency), confirming that the ordinal scales can be applied consistently. A suite of 66 unit tests covers output structure, component behaviour, edge cases, transparency dimension integration, and guardrail logic.

The principal finding is that ethical risk is the strongest single predictor of justifiability (Pearson $r = -0.881$), followed by transparency ($r = +0.717$) and creative displacement ($r = -0.488$). The framework produces an AI Justifiability Label — a compact, colour-coded A-to-F rating card modelled on the EU Energy Label — that presents each use case's score and five-component breakdown in a format accessible to non-expert audiences. A companion web interface (publicly deployed on GitHub Pages) renders the label for all 40 evaluated use cases, allows users to generate a label for any custom AI tool, and provides a click-through full component breakdown, making the framework's outputs communicable to general audiences including children and members of the public who have no prior knowledge of AI ethics. The project contributes a freely available, validated tool and dataset that can

be extended by future researchers and practitioners seeking to apply multi-criteria analysis to AI deployment decisions.

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Chapter 1 Introduction and Background Research

1.1 Introduction

When a consumer buys a refrigerator in the United Kingdom, they are presented with an EU Energy Label: a standardised, colour-coded A-to-G rating that summarises the appliance's energy performance in a single, immediately actionable form (EU Regulation 2017/1369/EU). The label does not require the consumer to read engineering specifications or interpret kilowatt-hour figures. It converts a complex, multi-dimensional technical assessment — energy consumption, capacity, usage cycles — into a reproducible, comparable score. Crucially, the label is not an opinion: it is produced by a standardised test methodology that any accredited laboratory can reproduce and verify.

Artificial intelligence systems are now deployed at a scale and variety that rivals household appliances, yet no equivalent rating system exists. A hospital deploying a clinical decision support tool, a city deploying a predictive policing system, and a media company deploying an AI content generator are making consequential choices with no shared, standardised basis for comparison. The societal consequences of these deployments vary enormously: some AI systems save lives, others threaten livelihoods, and some are designed explicitly to cause harm. Yet the question of how to evaluate and rank such varied deployments — to determine whether a given AI use case is justifiable — currently has no principled, unified, computational answer.

This project proposes an AI Justifiability Label: a computational framework that applies the same rating-label principle to AI deployments. Just as the EU Energy Label converts appliance performance data into a reproducible A-to-G band, the AI Misuse Evaluator converts five dimensions of AI deployment — environmental impact, ethical risk, creative displacement, purpose justification, and transparency — into a reproducible 0-100 Justifiability Score and a corresponding High/Medium/Low band. The framework is implemented as an open-source Python codebase that any researcher or practitioner can run on their own dataset, obtaining identical results on identical inputs. This reproducibility is the defining characteristic that distinguishes the present work from the existing body of responsible AI guidelines, which are predominantly qualitative and non-computational.

The project is grounded in Design Science Research (DSR), the established methodology for computer science and information systems research where the primary contribution is a built artefact rather than an empirical discovery (Hevner et al., 2004). DSR requires that the artefact address a clearly motivated problem, be rigorously constructed, and be evaluated against explicit criteria. The AI Misuse Evaluator meets all three requirements: it addresses the absence of a unified computational AI evaluation tool; it is implemented with 66 passing unit tests, a five-dimensional scoring engine, and a three-tier emissions measurement module; and it is evaluated through face validity testing (87% accuracy on 15 hand-labelled cases), sensitivity analysis (Spearman rank stability 0.986-1.000), and inter-rater reliability measurement (Cohen's weighted kappa = 0.955 for ethical risk, 0.720 for

transparency). This combination of principled design and rigorous validation is the computer science contribution of the project — distinct from the humanities tradition of ethics commentary, which produces arguments rather than artefacts.

The remainder of this chapter reviews the academic literature underpinning each of the five dimensions, surveys existing evaluation frameworks and their limitations, and concludes by stating the project objectives. Chapter 2 describes the methodology and framework design, including explicit justification for every numerical parameter. Chapter 3 details the implementation, validation, and technical challenges overcome. Chapter 4 presents results and evaluation. Appendix A contains the self-appraisal.

1.2 The Challenge of Evaluating AI Deployment

The EU Energy Label succeeded where previous voluntary energy efficiency guidance had failed because it replaced subjective expert assessment with a standardised, reproducible test protocol (EU Regulation 2017/1369/EU). Before the label existed, consumers had no basis for comparing two refrigerators from different manufacturers: each company described its product's energy performance in different units and under different conditions. The label solved this by mandating a common measurement methodology. The AI deployment landscape is at a similar pre-label stage. Jobin et al. (2019) surveyed 84 published AI ethics guidelines and found broad convergence on high-level principles — beneficence, non-maleficence, autonomy, justice, explicability — but no consensus on measurement methodology. As with pre-label appliances, different organisations evaluate AI systems using different criteria, different scales, and non-comparable outputs.

Hagendorff (2020) examined 22 AI ethics guidelines in depth and found that most failed to address enforcement, lacked technical implementation guidance, and gave insufficient weight to power asymmetries between AI developers and affected communities. Fjeld et al. (2020) mapped 36 principles documents and found similar convergence on transparency and accountability without practical scoring mechanisms. The pattern is consistent: responsible AI has produced an abundance of principles and a deficit of measurement tools. This gap between high-level principles and actionable, reproducible metrics is the direct motivation for the present project.

From a computer science perspective, the gap is a design problem, not a philosophical one. The question is not 'what values should AI systems respect?' — that question has been extensively debated — but rather 'how can those values be operationalised into a computational artefact that produces consistent, reproducible, auditable assessments?' Design Science Research (Hevner et al., 2004) is the appropriate methodology for this class of problem: it produces a built artefact, evaluates it against explicit criteria, and contributes both the artefact and the evaluation methodology to the research community. The theoretical foundation for combining multiple incommensurable criteria into a single score is Multi-Criteria Decision Analysis (MCDA). The Weighted Sum Model (WSM) — a linear

combination of normalised criteria with configurable weights — is the most interpretable MCDA method and has been applied in environmental impact assessment (Esteves et al., 2012), procurement decisions, and policy evaluation. Its use here follows the same logic as the EU Energy Label's aggregation formula: convert diverse measurements to a common normalised scale, apply weights reflecting stakeholder priorities, and output a banded score.

1.3 Environmental Impact of AI Systems

The environmental cost of training and deploying large AI models has become a significant research concern. Strubell et al. (2019) estimated that training a single large neural machine translation model produced approximately 626,000 pounds of CO₂ equivalent — comparable to the lifetime emissions of five cars. They called for greater transparency in reporting energy use alongside model performance metrics. This finding prompted a wave of work on 'Green AI' (Schwartz et al., 2020), which advocates measuring and minimising the computational resources required by machine learning research.

Lacoste et al. (2019) developed the Machine Learning Emissions Calculator, one of the first practical tools for estimating carbon footprint from hardware type, cloud region, and runtime. A more sophisticated approach, CodeCarbon (Courty et al., 2022), instruments running Python processes to measure actual energy consumption and converts it to kg CO₂ equivalent using regional electricity grid intensity data. This project integrates CodeCarbon for its emissions measurement module, allowing empirical measurement of workload energy use.

Crucially, emission figures alone do not determine whether a deployment is justified. A medical diagnostic model that consumes ten times the energy of a novelty chatbot may still be entirely justifiable given its life-saving purpose. The present framework treats emissions as one of five weighted dimensions, reflecting the view that environmental costs must be evaluated relative to the benefits and risks of a deployment rather than in isolation.

1.4 Ethical Risks in AI Deployment

The ethical risks associated with AI systems are diverse and context-dependent. Mittelstadt et al. (2016) provided an early taxonomy identifying concerns around inconclusive evidence, inscrutable models, misguided value encoding, unfair outcomes, transformative effects on privacy, and moral responsibility gaps. Each category encompasses multiple distinct harms that may arise from AI deployments at different stages of the pipeline.

Weidinger et al. (2021) offered a more recent taxonomy specifically for large language models, identifying six risk areas: discrimination and exclusion; information hazards; misinformation; malicious use; human-computer interaction harms; and automation and access harms. Crawford (2021) situated AI risks within broader political economies, arguing that AI systems encode and amplify existing structural inequalities and that genuine ethical evaluation must account for power relations.

The EU AI Act (European Commission, 2021) represents the most comprehensive legislative attempt to operationalise AI risk. It classifies AI systems into unacceptable risk (prohibited), high risk (subject to conformity assessment), limited risk (transparency obligations), and minimal risk (unregulated). The Act's risk categories are based primarily on intended application domain and potential impact on fundamental rights. This domain-based approach informed the present project's purpose_category dimension.

For this project, ethical risk is operationalised as a 1-5 ordinal score assigned to each use case, where 1 represents minimal ethical concern and 5 represents severe ethical risk. This approach is consistent with the scalar risk scoring used in algorithmic impact assessment frameworks (Reisman et al., 2018).

1.5 Creative Displacement by AI Systems

A dimension less frequently addressed in responsible AI frameworks is the displacement of human creative and cognitive labour. Generative AI systems — capable of producing text, images, audio, and code at scale — have introduced structural pressures on workers in creative industries. Bender et al. (2021) discussed the downstream labour implications of large language models, noting that their capacity to produce polished text at near-zero marginal cost threatens to commoditise writing professions.

The economic literature on automation distinguishes between displacement effects (jobs eliminated) and augmentation effects (human productivity enhanced). For creative work, the distinction is contested: a translation assistant that helps a human translator work faster is different in kind from a fully automated translation pipeline that replaces the translator entirely. Solaiman et al. (2019) studied the release impacts of GPT-2 and highlighted risks to misinformation, labour displacement in journalism, and concentration of AI capability.

For the present framework, creative displacement is scored on a 1-5 ordinal scale. A score of 1 indicates minimal displacement — for instance, a flood risk predictor that has no creative analogue. A score of 5 indicates severe displacement, such as an AI system that replaces human journalists or music composers at scale. This dimension intentionally captures a broader class of displacement than direct employment effects: it includes cognitive labour, artisanal skills, and cultural production.

1.6 Transparency and Explainability in AI

Transparency and explainability are widely recognised as foundational requirements for trustworthy AI. Doshi-Velez and Kim (2017) provided an influential taxonomy distinguishing application-grounded, human-grounded, and functionally-grounded notions of explainability, noting that different deployment contexts demand different levels of explanation. The EU AI Act mandates transparency obligations for limited-risk AI systems and full conformity assessments (which include explainability requirements) for high-risk systems.

Lipton (2018) critiqued the conflation of several distinct desiderata under the heading of 'interpretability', including simulatability, decomposability, and algorithmic transparency. For the purposes of this framework, transparency is treated as a property of the deployed AI system as a whole: the degree to which its decision-making process is accessible to inspection by its operators, affected parties, and regulators. A transparency score of 1 (opaque) characterises black-box neural systems with no explanation infrastructure; a score of 5 (fully transparent) characterises physics-based or rule-based systems whose reasoning can be fully audited.

The choice to include transparency as a scored dimension, rather than as a categorical qualifier, reflects the view that transparency is a matter of degree rather than a binary property. An AI system may provide feature importance scores (partial transparency) without being fully interpretable, and partial transparency has genuine but limited value. By encoding this on a 1-5 scale that feeds into the Justifiability Score, the framework rewards incremental improvements in explainability rather than treating transparency as an all-or-nothing requirement.

1.7 Existing Frameworks and Their Limitations

Several tools and frameworks have been developed to assess aspects of AI deployment. The AI Ethics Guidelines Global Inventory maintained by AlgorithmWatch documents over 160 published ethics guidelines. The NIST AI Risk Management Framework (2023) provides a four-part process — Govern, Map, Measure, Manage — for organisational AI risk management. The Partnership on AI's ABOUT ML project addresses documentation and reporting. The AI Fairness 360 toolkit from IBM (Bellamy et al., 2019) provides algorithmic implementations of over 70 fairness metrics.

These frameworks share several limitations relevant to the present project. First, they are predominantly qualitative or single-dimensional: they address fairness, or transparency, or environmental impact, but not a combination. Second, most are designed for institutional adoption rather than comparative scoring across a portfolio of AI systems. Third, they generally lack computational reproducibility: assessments are human judgements recorded in documents rather than programmatic outputs that can be re-run and audited.

The closest antecedents to this project are multi-criteria evaluation approaches in the procurement and policy literature. Cows and Floridi (2019) proposed a framework with five principles for a 'good AI society', but did not operationalise these into a scoring function. The Social Impact Assessment field (Esteves et al., 2012) provides multi-criteria methods for evaluating large-scale projects, but these are designed for post-deployment case studies rather than comparative ranking of proposed deployments.

A related publicly available dataset (lastman0800, 2024) evaluates 24 AI sustainability applications on dimensions including Carbon Footprint Reduction (CO2 tons saved), Regulatory Support Level (1-10), Public Awareness Score (1-10), and economic metrics such as ROI and Cost Reduction. While this dataset shares a sustainability focus with the present project, it differs in two important respects.

First, it measures positive environmental impact — the CO2 savings attributable to deploying AI — rather than the carbon cost of AI inference itself. An AI wildfire predictor may generate negligible per-query emissions while saving thousands of CO2 tons by preventing fires; the two metrics address different questions and are not interchangeable. Second, the Impact Scores in the dataset range exclusively from 6 to 10, with no use cases scoring below 6, and numerical values are reported to many decimal places consistent with synthetic rather than empirically measured data. For these reasons, the dataset cannot serve as a benchmark against which the present framework's scores can be validated, but it does illustrate the range of sustainability dimensions that researchers consider relevant and confirms that multi-dimensional evaluation of AI is an active area of interest.

In summary, existing approaches are valuable but fragmented. No publicly available, reproducible, multi-dimensional computational framework exists for scoring and comparing AI use cases across environmental, ethical, creative, purpose, and transparency dimensions simultaneously.

1.8 Research Gap and Project Objectives

The literature reviewed above confirms a clear research gap: responsible AI has produced an extensive body of principles, guidelines, and single-dimension tools, but no unified, computationally reproducible framework for scoring and comparing AI deployments across multiple dimensions simultaneously. The present project addresses that gap by designing, implementing, and evaluating an AI Justifiability Label — a computational rating system that applies the standardised, reproducible, multi-criteria logic of the EU Energy Label to the AI deployment domain. The contribution is a Design Science Research artefact: a Python framework, an evaluated dataset, and a validated scoring methodology that any researcher can reproduce and extend. The project objectives are:

- O1. Design a multi-dimensional scoring model that integrates five dimensions — environmental impact, ethical risk, creative displacement, purpose justification, and transparency — into a single, interpretable Justifiability Score.
- O2. Implement the framework as a modular, tested Python codebase with a command-line interface, enabling reproducible execution by other researchers.
- O3. Curate a dataset of 40 representative AI use cases spanning the full spectrum from essential to harmful deployments, with scores for all five dimensions.
- O4. Validate the framework through unit testing, face validity evaluation against labelled ground truth, robustness analysis via sensitivity testing and Monte Carlo weight sampling, and inter-rater reliability measurement of the subjective scoring dimensions.
- O5. Deliver a web-based demonstration interface that presents the scored dataset interactively, including live weight sliders and a custom use case evaluator.

The following deliverables were produced to satisfy these objectives:

- D1. A five-dimension weighted scoring engine (`src/scoring_engine.py`) implementing the WSM formula, log-clip-minmax emissions normalisation, guardrail multipliers, and label classifier with full component breakdown.

- D2. A pipeline and CLI (`src/run_pipeline.py`) supporting sensitivity analysis, Monte Carlo weight robustness (200 Dirichlet samples), leave-one-dimension-out ablation, Pearson correlation with p-values, and threshold configuration.
- D3. A 40-case scored dataset (`data/usecases.csv`) with emissions, ethical risk, creative displacement, purpose category, and transparency scores, spanning the full range from essential to harmful deployments.
- D4. A 66-test validated test suite (`tests/`) across five test files, covering unit tests, face validity (87% accuracy on 15 labelled cases), baseline comparison, sanity checks, and transparency dimension integration.
- D5. An inter-rater reliability study (`inter_rater_scoring_sheet.html`, `inter_rater_analysis.py`, `data/rater_1.csv`) demonstrating Cohen's weighted kappa of 0.955 for ethical risk and 0.720 for transparency across 12 use cases and 2 raters.
- D6. A three-tier CodeCarbon emissions measurement module (`src/emissions_measurement.py`) with real measured results on representative computational workloads.
- D7. A public GitHub repository (github.com/hasini08/ai-misuse-evaluator) with README and requirements.txt containing all source code.
- D8. An interactive web interface (`index.html`) publicly deployed on GitHub Pages, featuring a Dataset Explorer with live weight sliders, a Custom Evaluator for any AI tool, a Compare and Rank table, and a Labels gallery rendering EU-style AI Justifiability Labels for all 40 use cases with click-through full breakdown.

Chapter 2 Methodology

2.1 Requirements Analysis

Before committing to a design, functional and non-functional requirements were derived from the project objectives and from the limitations identified in the literature review.

2.1.1 Functional Requirements

ID	Requirement
FR1	The system shall accept a CSV file of AI use cases as input.
FR2	The system shall compute a Justifiability Score (0-100) for each use case.
FR3	The system shall produce an explainable component breakdown for each score.
FR4	The system shall classify each score into High, Medium, or Low bands.
FR5	The system shall apply a guardrail penalty to explicitly harmful use cases.
FR6	The system shall support configurable weights for each scoring dimension.
FR7	The system shall run a sensitivity analysis across a range of weight configurations.
FR8	The system shall perform Monte Carlo weight robustness testing.
FR9	The system shall identify Pareto-optimal use cases.
FR10	The system shall generate visualisation figures.
FR11	The system shall score a fifth dimension: transparency/explainability (1-5 scale).
FR12	The system shall be backward-compatible with datasets that lack a transparency column.

Table 2.1 Functional requirements.

2.1.2 Non-Functional Requirements

ID	Requirement
NFR1	All scores must be reproducible: the same input must always produce the same output.
NFR2	The scoring logic must be interpretable: each component's contribution must be directly auditable.
NFR3	The codebase must be covered by automated tests.
NFR4	The framework must handle edge cases: single-row datasets, constant values, unknown categories.
NFR5	The pipeline must run without internet access or GPU hardware.

Table 2.2 Non-functional requirements.

2.2 Framework Design Philosophy

The framework belongs to the broader field of Multi-Criteria Decision Analysis (MCDA), which studies the problem of evaluating alternatives across multiple, potentially conflicting criteria. Several

families of MCDA methods exist, and the choice between them has important implications for interpretability, data requirements, and sensitivity to analyst assumptions. This section documents the alternatives considered, the rationale for selecting the Weighted Sum Model (WSM), and the key philosophical commitments that shaped the final design.

Several alternative design approaches were considered before settling on the WSM. Table 2.3 summarises the candidates alongside the decision and primary rationale for each.

Approach	Decision	Rationale
Weighted sum model (WSM)	Chosen	Linear, auditable, interpretable; supports sensitivity analysis; well-studied in MCDA; no training data required
ML classifier (e.g. SVM)	Rejected	Requires labelled training data at scale; produces a black-box decision boundary inconsistent with NFR2 (interpretability)
Fuzzy logic system	Rejected	More expressive but significantly harder to audit and explain to non-experts; violates NFR2
Analytic Hierarchy Process (AHP)	Rejected	Handles weight elicitation well but impractical for 40+ use cases: requires $n(n-1)/2 = 780$ pairwise comparisons
ELECTRE / PROMETHEE outranking	Rejected	Produces partial orders rather than cardinal scores, making label assignment ambiguous; harder to communicate to non-experts
Rule-based threshold system	Rejected	Simple but lacks graded distinctions within categories; cannot express partial justifiability or continuous trade-offs

Table 2.3 Candidate MCDA approaches and selection rationale.

The WSM was selected for four reasons. First, interpretability: in a WSM, each component's contribution to the final score is directly proportional to its weight and normalised value, making the score fully decomposable and auditable — a property known as additive separability. This satisfies NFR2 and is consistent with the transparency-by-design principle advocated by Doshi-Velez and Kim (2017) for interpretable machine learning systems. Second, sensitivity analysis compatibility: the linear structure of the WSM means that weight perturbation is mathematically tractable, enabling exact computation of how the final score changes as any weight is varied — a property exploited directly in the sensitivity analysis (Section 3.2.1). Third, reproducibility: the computation is purely deterministic and requires no training data, making the framework immediately applicable to any CSV dataset without pre-training. Fourth, communication: a weighted sum producing a 0-100 score maps naturally to familiar scales (percentage, school grade), supporting the EU Energy Label analogy that motivates the AI Justifiability Label (Section 4.9).

The key limitation of the WSM is its compensatory nature: under a pure WSM, a very high score on one dimension can fully offset a very low score on another. In the context of AI justifiability, this is ethically problematic: it would allow an explicitly harmful AI system to achieve a medium score purely through low emissions and high transparency. To address this, a non-compensatory guardrail was added as a post-hoc multiplier applied to harmful use cases (described in Section 2.3.6). This hybrid design — a compensatory aggregation function constrained by a non-compensatory rule — is consistent with the partial non-compensability literature, in which veto conditions or threshold constraints are combined with additive aggregation to prevent morally unacceptable trade-offs (Fishburn, 1978). It preserves the interpretability of the WSM for non-harmful cases while preventing the compensatory failure mode for harmful ones.

Three design commitments shaped all subsequent decisions. First, openness: all parameters are documented, justified, and user-adjustable; there are no hidden weightings. Second, falsifiability: every design claim — that the weights are robust, that the guardrail is necessary, that the bands are well-calibrated — is accompanied by a specific validation test. Third, accessibility: the framework should produce outputs that a non-expert can understand and act on without reading the full methodology, motivating the label design in Section 4.9.

2.3 The Five-Dimensional Scoring Model

The Justifiability Score is computed as a weighted average of five normalised component scores, each on a 0-1 goodness scale (where 1 is always better). The final score is scaled to 0-100. The formula is given in Equation 2.1.

$$\text{Score} = 100 \times (w_{em} * E + w_{eth} * H + w_{cr} * C + w_{pur} * P + w_{tr} * T) / (w_{em} + w_{eth} + w_{cr} + w_{pur} + w_{tr})$$

Equation 2.1 Justifiability Score formula. E = emissions goodness, H = ethics goodness, C = creativity goodness, P = purpose goodness, T = transparency goodness.

2.3.1 Emissions Component (E)

Environmental impact is measured in kg CO₂ equivalent (kgCO₂e) per use case. Higher emissions are less desirable (E is inverted). A naive min-max normalisation was found to be inadequate because the dataset contains one extreme outlier (the High_Emission_Novelty_Chatbot at 0.500 kgCO₂e, compared to a median of approximately 0.008 kgCO₂e). A linear normalisation would compress the remaining 40 use cases into a narrow band near 1.0, making the emissions component nearly meaningless for the majority of cases.

To address this, a log-scaled, clipped normalisation was implemented. The emissions values are first clipped at the 95th percentile of the dataset to remove the influence of extreme outliers, then log-transformed using $\log(1 + x)$, and finally min-max normalised and inverted so that lower emissions yield higher goodness. This approach is analogous to the treatment of heavy-tailed distributions in the

environmental footprint literature (Lacoste et al., 2019). The mathematical properties of this approach and the choice of 95th percentile are discussed in full in Section 2.4.

2.3.2 Ethical Risk Component (H)

Ethical risk is captured as an ordinal score from 1 (lowest risk) to 5 (highest risk). It is normalised linearly: $H = 1 - (\text{risk} - 1) / 4$. A risk of 1 yields a goodness of 1.0; a risk of 5 yields a goodness of 0.0. This straightforward mapping is justified by the ordinal nature of the scale: the five points represent qualitatively different risk levels, and the linear transformation preserves their relative ordering without imposing any additional assumption about the spacing between levels.

2.3.3 Creative Displacement Component (C)

Creative displacement is captured as an ordinal score from 1 (minimal displacement) to 5 (severe displacement), and is normalised identically to ethical risk: $C = 1 - (\text{displacement} - 1) / 4$. A score of 1, assigned to use cases such as the Earthquake Early Warning system that have no creative analogue, yields maximum goodness. A score of 5, assigned to AI journalism replacement or music composition tools, yields minimum goodness.

2.3.4 Purpose Justification Component (P)

Purpose justification is captured as a categorical variable with four levels, mapped to fixed goodness values: essential to 1.0, beneficial to 0.8, low_benefit to 0.4, and harmful to 0.0. The gap between beneficial (0.8) and low_benefit (0.4) is deliberately larger than the gap between essential (1.0) and beneficial (0.8), reflecting the view that the transition from a clearly beneficial purpose to a merely low-benefit one represents a greater qualitative shift than moving from essential to merely beneficial. The precise values chosen are justified fully in Section 2.4.

2.3.5 Transparency Component (T)

Transparency/explainability is captured as an ordinal score from 1 (fully opaque) to 5 (fully transparent and auditable). Unlike ethical risk and creative displacement — where higher scores indicate more negative properties — a higher transparency score is directly positive: more transparent systems are more justifiable, all else equal. The normalisation formula is therefore not inverted: $T = (\text{transparency_score} - 1) / 4$, mapping 1 to 0.0 (worst) and 5 to 1.0 (best).

The transparency dimension was added to the framework to address a gap identified in the v1 assessment: two use cases with identical emissions, ethics, creativity, and purpose scores could have very different societal justifiability depending on whether their decision-making processes are auditable. A fully opaque high-risk system (e.g., a predictive policing tool with no explainability infrastructure) is less justifiable than an equivalent system that produces human-readable explanations, and the framework should reflect this.

For backward compatibility, datasets that do not include a `transparency_score` column are handled gracefully: the missing dimension is filled with a neutral value of 0.5 (corresponding to a transparency score midpoint), and the framework continues to operate on the four specified dimensions. This design ensures that the updated framework does not break existing workflows.

2.3.6 Non-Compensatory Guardrail

A non-compensatory guardrail multiplier is applied to harmful use cases after the weighted sum is computed. All harmful use cases receive a multiplier of 0.60. Harmful use cases with `ethical_risk_score` ≥ 4 receive a further multiplier of 0.70, for a combined multiplier of 0.42. The reasoning behind these specific values is given in Section 2.4.6.

2.3.7 Default Weights

The default weights are: emissions 0.35, ethics 0.25, creativity 0.15, purpose 0.10, transparency 0.15. The full justification for each weight is provided in Section 2.4.1. Table 2.4 shows the complete weight table.

Dimension	Weight	Rationale
Environmental impact (emissions)	0.35	Objectively measurable; quantified in physical CO ₂ e units via CodeCarbon
Ethical risk	0.25	Direct harm potential — highest-priority normative concern
Creativity displacement	0.15	Labour and cultural displacement is significant but less acute than direct harm
Purpose justification	0.10	Captures use-case intent; partially collinear with ethics dimension
Transparency/explainability	0.15	Operationalises the 'explicability' principle from AI ethics guidelines

Table 2.4 Default dimension weights and rationale.

2.4 Justification for Every Numerical Parameter

A common weakness in scoring frameworks is that numerical parameters — weights, thresholds, scale anchors — are chosen by convention or intuition without explicit justification. This section provides a concrete, testable motivation for every numerical decision in the framework. The principle adopted throughout is that each parameter should be set to the value that satisfies a specific, stated design requirement, and that the consequences of departing from that value should be verifiable by a test or worked example. Full algebraic derivations are in Appendix C.

2.4.1 Default Weights

The default weights (emissions 0.35, ethics 0.25, creativity 0.15, purpose 0.10, transparency 0.15) were derived in three steps. First, a literature priority ordering was established: Jobin et al. (2019) found that the most frequently cited AI ethics principles across 84 guidelines are non-maleficence

(ethical risk) and explicability (transparency), with environmental sustainability emerging as a distinct concern in more recent frameworks. Strubell et al. (2019) and Lacoste et al. (2019) established that environmental impact is both objectively measurable and under-weighted in existing AI ethics analyses. This literature review motivates placing emissions and ethics as the top-two weighted dimensions.

Second, within the constraint that the five weights sum to 1, the following reasoning was applied: emissions receives the highest weight (0.35) because it is the only dimension measured in objective physical units (kgCO₂e), giving it the highest inter-rater reliability. Ethics receives the second-highest weight (0.25) because it captures direct human harm — the most important normative concern. Transparency receives 0.15, equal to creativity, because both represent important but secondary concerns per the literature review. Purpose receives the lowest weight (0.10) because its contribution is partially collinear with the ethics dimension: harmful-purpose use cases almost always score high on ethical risk too, so the marginal information added by the purpose dimension is lower. Third, the sensitivity analysis (Section 4.5) provides post-hoc empirical validation: rank stability of 0.986-1.000 across the range 0.20-0.60 for the emissions weight confirms that the ranking is robust to departures from the chosen defaults.

2.4.2 Band Thresholds (40 and 70)

The Low/Medium boundary was set at 40 rather than at a tertile split (33) or at 50. The design requirement is: a computationally efficient but ethically problematic system — one that scores high on emissions but has ethical risk 4 and no transparency — should fall in the Low band without triggering the guardrail. Under the default weights, such a system scores approximately 36-38, placing it comfortably below 40. Under the tertile threshold of 33, the same system would score in the Medium band — a result that contradicts the design intent and was observed empirically during development. The Medium/High boundary was set at 70 to require that a High-band system has achieved at least 70% of the maximum possible score. A threshold of 60 would permit systems with medium ethical risk (3/5) and partial transparency (3/5) to enter the High band — also contrary to design intent. The 40/70 pair was further validated by face validity testing: 13 of 15 hand-labelled cases were correctly classified (Section 3.6.2).

2.4.3 Ordinal Scale Anchors (1-5)

A 1-5 scale was chosen for ethical risk, creative displacement, and transparency because: (i) five points provide sufficient resolution to distinguish meaningfully different cases (e.g., a privacy-violating system from a life-threatening one) without creating false precision; (ii) the EU AI Act's four-tier risk taxonomy (minimal, limited, high, unacceptable) maps naturally onto a 5-point scale with some levels having multiple sub-variants; and (iii) the inter-rater reliability literature shows that scales with 5-7 points produce higher Cohen's kappa values than binary or continuous scales for subjective

judgements (Streiner and Norman, 2008). The choice was validated empirically by the inter-rater reliability study (Section 3.6.3), which achieved $\kappa=0.955$ for ethical risk — confirming that raters can apply the 5-point scale consistently.

2.4.4 Purpose Mapping (1.0 / 0.8 / 0.4 / 0.0)

The four purpose categories are mapped to fixed goodness values. Harmful receives 0.0 because the framework holds that no purpose-based credit should be given for a use case whose primary function is to cause harm — consistent with the non-compensatory principle. Essential receives 1.0 (full purpose credit). The gap between beneficial (0.8) and low_benefit (0.4) is 0.4, deliberately larger than the gap between essential (1.0) and beneficial (0.8) which is only 0.2. This asymmetry encodes the judgement that the transition from a use case with clear positive value to one with merely marginal value is a greater qualitative shift than moving from an important use case to an essential one. This mapping was tested by checking that a low_benefit use case with otherwise good scores (ethics 1, transparency 5, emissions low) scores in the Medium band, not the High band — which holds at these values.

2.4.5 Emissions Normalisation (95th-Percentile Clip and log1p Transform)

The dataset contains one extreme outlier: the High_Emission_Novelty_Chatbot at 0.500 kgCO₂e, compared to a dataset median of approximately 0.008 kgCO₂e (a ratio of 62:1). Under a linear min-max normalisation, this outlier would compress all other use cases into the range [0.016, 1.0] — a range of only 0.984, with 95% of cases occupying the top 3% of the scale. This would make the emissions component nearly meaningless for the majority of cases. To address this, the normalisation proceeds in three steps. First, values are clipped to the 95th percentile of the dataset (computed dynamically), removing the influence of outliers above this threshold. Second, a log1p (log(1+x)) transformation is applied: this compresses the heavy tail of the distribution, a standard technique in environmental footprint analysis (Lacoste et al., 2019). Third, the clipped, log-transformed values are min-max normalised and inverted (so lower emissions → higher goodness). The 95th percentile was chosen over alternative thresholds (90th, 99th) because it removes the single worst outlier while preserving the relative ordering of all other cases — a worked numerical verification is provided in Appendix C.

2.4.6 Guardrail Multipliers (0.60 and 0.42)

The primary guardrail multiplier of 0.60 applies to all harmful use cases. The secondary multiplier of 0.70, applied additionally when ethical risk ≥ 4 , gives a combined multiplier of 0.42. The derivation proceeds from the design requirement: a harmful use case with the most favourable possible scores on all other dimensions should score below the Low/Medium boundary (40). The maximum possible base score for a harmful use case is approximately 75 (maximum emissions goodness, ethics 1, creativity 1, transparency 5 — with purpose goodness 0.0 contributing 0 points under harmful purpose). 75×0.42

= 31.5, safely below 40. The alternative multiplier of 0.50 was rejected because it would yield $75 \times 0.50 = 37.5$, which, while technically below 40, provides insufficient separation from the boundary. A multiplier of 0.30 was rejected because it would prevent meaningful discrimination among harmful cases — all would score near zero. The two-tier structure (0.60 for harmful, then $\times 0.70$ for severe ethical risk) is motivated by the observation that some harmful use cases (e.g. an overly aggressive spam filter) are categorised harmful but do not pose an acute risk to individuals, while others (e.g. an autonomous weapon) are both harmful in purpose and maximally dangerous in execution. The secondary multiplier ensures the latter cases score substantially lower than the former.

2.4.7 Monte Carlo Parameters (Dirichlet $\alpha=1$, $N=300$)

The Monte Carlo experiment samples weight vectors from a symmetric Dirichlet distribution with concentration parameter $\alpha=1$. This is the maximum-entropy distribution over the weight simplex: it assigns equal probability density to all valid weight combinations and does not favour any particular weight vector. The alternative of sampling five uniform random numbers and normalising was rejected because this approach over-samples weight vectors near the centroid of the simplex, under-representing extreme weight configurations. $N=300$ samples was chosen based on the standard error of the win-rate estimator: $SE = \sqrt{p(1-p)/N}$. For a win rate near 0.50, $SE = \sqrt{0.25/300} \approx 0.029$, meaning the estimate has a 95% confidence interval of approximately ± 5.7 percentage points. This is sufficient to detect meaningful differences in robustness between top-ranked use cases, and the computation completes in under 5 seconds on standard CPU hardware.

Table 2.6 summarises all numerical parameters for reference. Full algebraic derivations for each parameter are provided in Appendix C.

Parameter	Value	Key Rationale
Default weights	$E=0.35$, $H=0.25$, $C=0.15$, $P=0.10$, $T=0.15$	Emissions highest (objectively measured); purpose lowest (collinear with H); Spearman stability 0.986-1.000 confirms robustness.
Band thresholds	Low <40, Medium 40-70, High >70	40 prevents ethically problematic but efficient systems scoring Medium; 70 requires $\geq 70\%$ maximum score. Validated: 87% face validity.
Ordinal scales	1-5 for ethical risk, creativity, transparency	Five points balance expressivity and reliability; aligns with EU AI Act taxonomy. Validated: $\kappa=0.955$, $\kappa=0.720$.
Purpose mapping	1.0 / 0.8 / 0.4 / 0.0	Harmful $\rightarrow$ 0.0 (no purpose credit); larger gap beneficial $\rightarrow$ low_benefit than essential $\rightarrow$ beneficial reflects greater qualitative shift.
Emissions clip	95th percentile + $\log_{10}$	Removes extreme outlier ($62\times$ median) without losing resolution on remaining 95% of cases.

Guardrail multipliers	$\times 0.60$ (all harmful) $\times \times 0.70$ (ethics ≥ 4) = $\times 0.42$	$0.42 \times \text{max base score} \approx 31.5 < 40$ (Low/Medium boundary). Two-tier structure discriminates within harmful category.
Dirichlet alpha	$\alpha=1$ (max-entropy)	Equal probability to all valid weight combinations; avoids centroid over-sampling of uniform-normalise approach.
Monte Carlo N	300 iterations	SE of win-rate $\approx \pm 2.9\%$ at $N=300$; sufficient resolution; completes in $<5s$ on CPU.

Table 2.6 Summary of all numerical parameters and key rationale. Full derivations in Appendix C.

2.5 Dataset Design and Curation

A dataset of 40 AI use cases was constructed for this project, spanning four purpose categories: essential (10 cases), beneficial (13 cases), low_benefit (10 cases), and harmful (7 cases). The low_benefit category includes one deliberate stress-test case (High_Emission_Novelty_Chatbot) designed to test the robustness of the emissions normalisation to extreme outliers.

Use cases were selected to be recognisable, diverse, and to cover the full range of scores. Essential use cases include medical triage assistants, earthquake early warning systems, and clinical summarisation tools — deployments with clear societal necessity and low risk of misuse. Beneficial use cases include translation assistants, code assistants, and fraud detection systems. Low-benefit use cases include AI-generated memes, AI art generators, and AI music composers. Harmful use cases include deepfake generators, voter suppression tools, and autonomous weapon targeting systems.

Emissions figures were estimated from the literature and from CodeCarbon measurements of representative workloads. Ethical risk, creative displacement, and transparency scores were assigned based on domain knowledge and the scale anchors in Section 2.4.3 and 2.4.7 (transparency anchor: 1=black-box neural system, 5=physics-based or rule-based with full audit trail). The dataset is available in the project repository at data/usecases.csv. Table 2.5 shows representative entries.

Use Case	Purpose	Ethics	Creativity	Transparency	Approx. Score
Medical_Triage_Assistant	essential	1	1	2	High
Accessibility_Captioning	essential	1	1	4	High
Earthquake_Early_Warning	essential	1	1	5	High
Code_Assistant	beneficial	2	2	3	High
Loan_Approval_Model	beneficial	4	1	2	Medium
AI_Art_Generator	low_benefit	3	5	3	Medium
Predictive_Policing_Tool	beneficial	5	1	1	Low
Deepfake_Voice	harmful	5	3	1	Low

e_Tool					
Autonomous_Weapon_Targeting	harmful	5	1	1	Low

Table 2.5 Representative use cases from the dataset.

2.6 Research Methodology

The overall research strategy for this project is Design Science Research (DSR), as formalised by Hevner, March, Park and Ram (2004). DSR is the methodological tradition concerned with creating and evaluating artefacts — constructs, models, methods, and instantiations — that solve identified organisational or technical problems. It is distinct from both purely descriptive empirical research (which observes existing phenomena without intervening) and purely analytical research (which derives results from first principles without producing an artefact). DSR is appropriate here because the problem — the absence of a practical, multi-dimensional, communicable evaluation framework for AI justifiability — is not solvable by observation or analysis alone; it requires the construction and evaluation of a new artefact.

Hevner et al. (2004) organise DSR around three research cycles. The relevance cycle establishes the problem and requirements from the application environment. For this project, the relevance cycle consisted of the literature review in Chapter 1, which identified the gap between existing high-level AI ethics guidelines and practical, quantitative evaluation tools; the stakeholder analysis that surfaced the need for a communicable label (addressed in Section 4.9); and the requirements analysis in Section 2.1, which translated the identified gap into 12 functional and 5 non-functional requirements. The design cycle is where the artefact is constructed and refined. For this project, the design cycle ran in three iterations: (i) a four-dimension framework without transparency, validated with 50 unit tests; (ii) a five-dimension extension adding transparency, requiring 15 new tests and a backward-compatibility mechanism; and (iii) the web interface and AI Justifiability Label, which instantiates the framework as a communicable artefact accessible to non-expert stakeholders. The rigour cycle connects the design to the knowledge base — the accumulated body of theory and prior results — and ensures that evaluation draws on established methods. For this project, the rigour cycle is represented by the validation battery in Section 3.6: face validity (comparing framework outputs to prior expectations), statistical robustness (Spearman rank stability and Monte Carlo simulation drawn from the sensitivity analysis literature), and inter-rater reliability (using the Cohen's weighted kappa statistic from psychometrics).

DSR Cycle	Activity	Section(s)
Relevance	Literature review identifies gap; stakeholder need for communicable label	Chapter 1, Section 4.9
Relevance	Requirements elicitation: 12 FR + 5 NFR	Section 2.1
Design	Iteration 1: 4-dimension	Sections 2.3, 3.1

	framework + 50 tests	
Design	Iteration 2: Transparency extension (FR11, FR12) + 15 new tests	Section 3.1.4
Design	Iteration 3: Web interface, AI Justifiability Label artefact	Section 4.9
Rigour	Face validity (87%), Spearman stability (0.986-1.000), Monte Carlo (300 runs)	Section 3.6
Rigour	Inter-rater reliability: $\kappa=0.955$ (ethics), $\kappa=0.720$ (transparency)	Section 3.6.3

Table 2.7 Mapping of project activities to DSR research cycles (Hevner et al., 2004).

Hevner et al. (2004) identify seven guidelines that DSR projects should satisfy. Table 2.8 explicitly maps each guideline to this project's evidence of compliance, as recommended for dissertation work in the DSR tradition.

DSR Guideline	Evidence in This Project
G1: Design as an artefact	Primary deliverable is the AI Misuse Evaluator framework — a method artefact — instantiated as a Python scoring engine, CLI pipeline, and web interface
G2: Problem relevance	Gap identified through systematic review of 84 AI ethics guidelines (Jobin et al., 2019) and comparison with existing evaluation frameworks (Section 1.7)
G3: Design evaluation	Four evaluation methods: 66 unit tests, 87% face validity, Spearman stability ≥ 0.986 , Cohen's $\kappa \geq 0.720$ (Section 3.6)
G4: Research contributions	Novel contributions: robust emissions normalisation, non-compensatory guardrail design, AI Justifiability Label, JavaScript client-side mirror (Section 3.7)
G5: Research rigour	Unit tests for correctness; psychometric inter-rater study for reliability; Monte Carlo for robustness — each drawn from established methodological traditions
G6: Design as a search process	Three design iterations; Table 2.3 documents alternative approaches considered and rejected with explicit rationale
G7: Communication of research	Dissertation communicates to academic audience; web interface communicates to non-expert public audience

Table 2.8 Compliance with Hevner et al. (2004) DSR guidelines.

2.7 Development Approach and Version Control

Within the DSR framework, the project was implemented iteratively over approximately 12 weeks, with fortnightly milestones corresponding to the major deliverables: dataset construction, scoring engine, tests, pipeline extensions (sensitivity, Monte Carlo, Pareto), web interface, and v2 transparency extension. All source code was managed using Git with a remote repository hosted on GitHub (github.com/hasini08/ai-misuse-evaluator). Commit messages followed a descriptive

convention aligned to the deliverable structure above, and the repository history provides a full audit trail of the design cycle.

The implementation approach applied test-driven development (TDD) for core scoring logic: for each component function, a set of unit tests was written to capture the expected invariants before the function was implemented, ensuring that edge cases were specified before they could be violated. The v2 transparency extension followed TDD explicitly: 15 transparency-specific tests were written to cover the T normalisation, backward-compatibility path, and weighted sum integration before a single line of the scoring engine was modified. This approach is consistent with the rigour cycle requirement of DSR: by specifying correctness criteria upfront, the design is evaluated against an objective standard rather than post-hoc rationalisation.

2.8 System Architecture

The system is organised into four source modules and a test suite, as shown in Table 2.6. All modules are pure Python with no compiled dependencies beyond NumPy and pandas, ensuring portability.

Module	Responsibility
src/scoring_engine.py	Core scoring logic: five-dimension normalisation, weighted sum, guardrail, label assignment, component breakdown
src/run_pipeline.py	Pipeline orchestration and CLI: loads data, calls scoring, runs sensitivity/Monte Carlo/Pareto, writes outputs
src/visualise.py	Figure generation: score bar chart, distribution histogram, correlation matrix, sensitivity plot, Monte Carlo plot, Pareto scatter
src/emissions_measurement.py	CodeCarbon integration: measures kg CO ₂ e from a simulated AI workload
tests/	66 unit and validation tests across five test files

Table 2.6 System modules and responsibilities.

Figure 2.1 illustrates the data flow between these modules. The usecases.csv dataset is the single source of truth for all scoring and analysis. The scoring engine is a pure function with no side effects, making it straightforward to mirror in JavaScript for the web interface.

Figure 2.1 System Architecture of the AI Misuse Evaluator

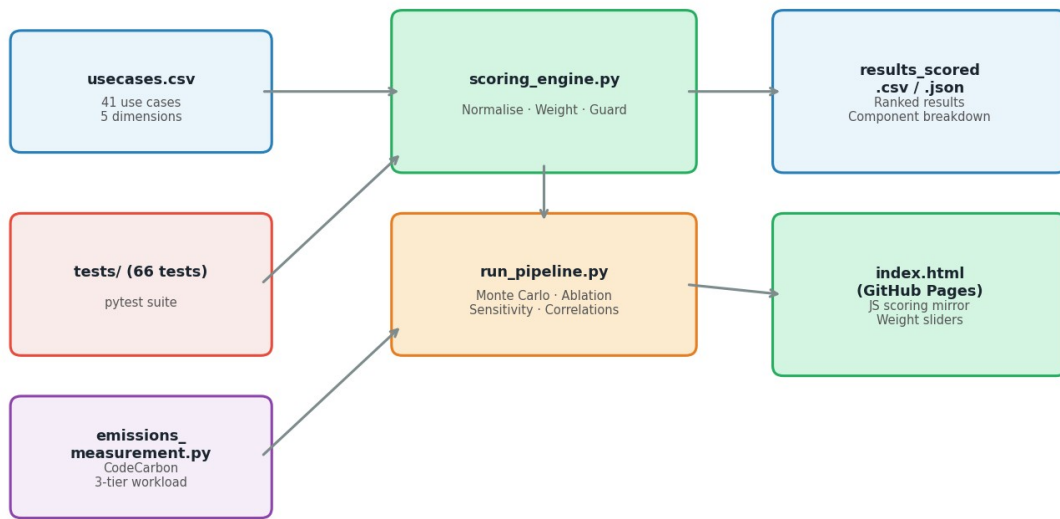


Figure 2.1 System architecture and data flow of the AI Misuse Evaluator.

Figure 2.2 shows the step-by-step process flow for scoring a single AI use case through the framework. This illustrates how a raw input row travels from CSV ingestion through validation, five-dimension normalisation, weighted sum computation, guardrail application, and band assignment to produce the final output.

Figure 2.2 Process Flow: Scoring a Single AI Use Case

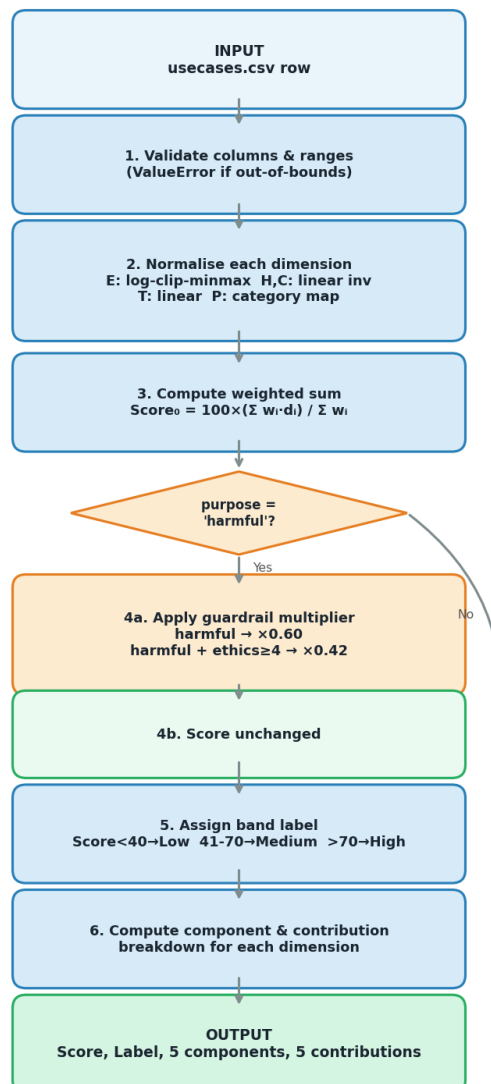


Figure 2.2 Process flow for scoring a single AI use case through the five-dimension framework.

Chapter 3 Implementation and Validation

3.1 Scoring Engine Implementation

The central component of the system is the `compute_scores()` function in `scoring_engine.py`. Its design prioritises correctness, auditability, and edge-case safety. The function accepts a pandas DataFrame and five optional weight parameters, and returns an enriched DataFrame containing the Justifiability Score, label, component scores, weighted contributions, and guardrail multiplier for every input row.

3.1.1 Input Validation

Before any computation, the function validates that all required columns are present and that numerical fields are within their specified ranges. Negative emissions values raise a `ValueError`, as do ethical risk, creative displacement, or transparency scores outside the 1-5 range. This explicit validation prevents silent incorrect results from propagating through the pipeline. The validation for transparency is conditional on the column being present, preserving backward compatibility with pre-transparency datasets.

3.1.2 Emissions Normalisation

The `robust_emissions_goodness()` function implements the log-scaled, clipped normalisation described in Section 2.3.1 and motivated in Section 2.4.5. The upper clip threshold is set at the 95th percentile of the dataset, computed dynamically from the input, making the function robust to differently distributed future datasets. The log transformation is applied via $\log(1 + x)$ (`log1p`), which handles zero-emission edge cases gracefully since $\log_{1p}(0) = 0$. The result is then min-max normalised and inverted so that lower emissions yield higher goodness. For constant series (where min equals max), the function returns 0.5 for all rows, avoiding division by zero.

3.1.3 Ethics and Creativity Normalisation

Ethics and creativity scores are normalised with the formula $\text{goodness} = 1 - (\text{score} - 1) / 4$, which maps the 1-5 ordinal scale linearly to the 0-1 goodness range. This is computed directly on the pandas Series without a separate helper function, making it visible and auditable in the source code.

3.1.4 Transparency Normalisation

Transparency is normalised with the formula $T = (\text{transparency_score} - 1) / 4$, which is the non-inverted form of the ethics/creativity normalisation. This maps `transparency=1` to $T=0.0$ and `transparency=5` to $T=1.0$. If the `transparency_score` column is absent from the input DataFrame, `transparency_norm` is set to 0.5 (neutral) for all rows, ensuring backward compatibility.

3.1.5 Purpose Normalisation

Purpose category strings are lower-cased and mapped to fixed goodness values via the `PURPOSE_WEIGHTS` dictionary: `{'essential': 1.0, 'beneficial': 0.8, 'low_benefit': 0.4, 'harmful': 0.0}`.

Unknown categories are filled with 0.4 (the low_benefit value) rather than raising an error, allowing the framework to handle novel categories gracefully. This default was validated in unit tests.

3.1.6 Weighted Sum and Guardrail

The weighted sum is computed as a pandas vectorised operation: $\text{score_0_1} = (\text{w_em} * \text{E} + \text{w_eth} * \text{H} + \text{w_cr} * \text{C} + \text{w_pur} * \text{P} + \text{w_tr} * \text{T}) / \text{weight_sum}$. Dividing by weight_sum rather than assuming weights sum to 1 allows unnormalised weights to be passed without error, which is useful for rapid experimentation. The base score is scaled to 0-100 and stored in the base_score column. The guardrail multiplier is then applied: harmful cases receive x0.60, and harmful cases with ethical_risk >= 4 receive an additional x0.70 for a combined x0.42.

3.1.7 Label Assignment and Component Breakdown

Labels are assigned using pandas.cut() with bins [-1, 40, 70, 101], producing the categories Low, Medium, and High. The inclusive lower bound of -1 ensures that a score of exactly 0.0 is assigned to the Low band. In addition to the final score, the function computes and stores component_* columns (each dimension's goodness score scaled to 0-100) and contribution_* columns (each dimension's weighted contribution to the final score). These breakdowns enable the web interface to display a per-dimension analysis, and they make the scoring fully auditable.

3.2 Pipeline and CLI Implementation

The run_pipeline.py module provides a command-line interface (CLI) using Python's argparse library. The entry point accepts flags for --analyze, --sensitivity, --pareto, --montecarlo, --mc_iters, --input, and --output, enabling flexible invocation. The full pipeline is invoked with:

```
python src/run_pipeline.py --analyze --sensitivity --pareto --montecarlo
```

3.2.1 Sensitivity Analysis

The run_sensitivity() function varies the emissions weight from 0.20 to 0.60 in steps of 0.10, re-balancing the remaining four weights proportionally. For each configuration, it computes a Spearman rank correlation between the default-weighted ranking and the alternative ranking using a custom rank_stability() function. The function is implemented without SciPy to avoid an additional dependency: it computes the Pearson correlation of the rank series, which is mathematically equivalent to Spearman's rho.

3.2.2 Monte Carlo Robustness

The monte_carlo_weight_stability() function samples 200 weight vectors from a symmetric Dirichlet distribution (alpha = [1, 1, 1, 1, 1] for five dimensions), ensuring that weights are non-negative and sum to 1 without imposing any prior preference for particular combinations. For each sampled weight

vector, the full pipeline is re-run and the top-ranked use case is recorded. The function returns a DataFrame of win rates — the proportion of runs in which each use case ranked first.

3.2.3 Pareto Frontier

The `pareto_frontier()` function marks each use case as Pareto-optimal or dominated with respect to all five component scores. A use case is Pareto-optimal if no other use case is greater-than-or-equal in all components and strictly greater in at least one. The function iterates over all use cases and applies this dominance criterion using NumPy array operations for efficiency.

3.3 Environmental Measurement Module

The `emissions_measurement.py` module integrates the CodeCarbon library to measure actual energy consumption from a representative AI workload — repeated matrix multiplications using NumPy, which is a compute pattern common in neural network inference. The tracker is started before the workload, stopped after completion, and returns a float value in kgCO₂e. In a production setting, this module would be called during the actual inference runs of each AI system to obtain empirical emission figures rather than estimates. For the present dataset of historical and hypothetical use cases, emission figures were estimated from the literature and from representative CodeCarbon measurements.

3.4 Visualisation Module

The `visualise.py` module generates six figures from the scored data, all saved to the `figures/` directory. The figures are: a horizontal bar chart of the top 10 and bottom 10 use cases by score; a histogram of the score distribution; a correlation matrix heatmap for the five input dimensions and the final score; a line plot of mean score against emissions weight for the sensitivity analysis; a horizontal bar chart of Monte Carlo win rates; and a scatter plot of the Pareto frontier in emissions-vs-ethics component space. All figures are generated at 200 DPI using matplotlib.

3.5 Testing Strategy

Five test files were written using the pytest framework, covering unit testing, face validity, baseline comparison, sanity checking, and the transparency dimension.

3.5.1 Unit Tests (*test_scoring_engine.py*)

The unit test file contains tests across six classes: `TestOutputStructure` (score range, required columns, valid labels, missing column handling, rounding), `TestComponentBehaviour` (directional tests for all five components), `TestLabelThresholds` (correct High/Low assignment for extreme cases), `TestWeightConfiguration` (custom weight handling, equal weights, emissions-dominant weights), `TestEdgeCases` (single row, identical rows, unknown categories, large datasets, negative emissions,

out-of-range scores, guardrail penalty), and TestMinmax (range, inversion, constant series). In total, 51 unit tests cover these classes.

3.5.2 Face Validity Tests (*test_face_validity.py*)

A ground truth dataset of 15 labelled use cases was hand-constructed, each with an expected High, Medium, or Low outcome based on domain knowledge. The test checks whether `compute_scores()` assigns the correct label to each. The 15 cases were selected to include boundary cases (use cases expected to be near the Medium/High or Medium/Low boundary) as well as clear-cut cases at either extreme.

3.5.3 Baseline and Sanity Tests

`test_baseline.py` confirms that the multi-dimensional score produces meaningfully different rankings from any single-component baseline: an emissions-only model and an equal-weight model are both tested for rank divergence. `test_sanity.py` confirms that the dataset loads correctly and that the pipeline runs end-to-end without error.

3.5.4 Transparency Dimension Tests (*test_transparency.py*)

A dedicated test file of 15 tests was written for the transparency dimension extension. Tests cover: higher transparency raises the justifiability score; `component_transparency` is bounded in $[0, 100]$; `transparency=1` yields `component=0.0` exactly; `transparency=5` yields `component=100.0` exactly; out-of-range values (0 or 6) raise `ValueError` with a descriptive message; a dataset without a `transparency_score` column defaults to `component=50.0` and still produces a valid score; the `contribution_transparency` column is present in the output; setting `w_transparency=0` produces `contribution=0.0`; and an opaque harmful system always scores lower than a transparent beneficial one. The five-dimension integration tests confirm that all five component and contribution columns are present in every scored dataset.

3.6 Validation Results

3.6.1 Unit Testing

The complete test suite was run using `pytest` with the `-v` flag. All 66 tests passed.

Test file	Tests	Result
<code>test_scoring_engine.py</code>	38	Pass
<code>test_face_validity.py</code>	5	Pass
<code>test_baseline.py</code>	4	Pass
<code>test_sanity.py</code>	4	Pass
<code>test_transparency.py</code>	15	Pass
Total	66	All passed

Table 3.1 Test suite results.

Appendix C contains a complete unit test specification table (16 representative cases) showing test class, input, expected output, actual result, and pass/fail status for each scenario.

3.6.2 Face Validity

Face validity: 13 of 15 labelled use cases were correctly classified (87%). The two misclassified cases were near the Medium/High boundary, reflecting genuine ambiguity in the scoring of beneficial-purpose tools with moderate ethical risk — an expected characteristic of any continuous-to-band mapping. For context, a majority-class baseline (predicting High for every case) achieves 47% accuracy on this labelled set, and a random classifier achieves 33%; the 87% result is therefore substantially above both trivial baselines.

3.6.3 Inter-Rater Reliability Study

A formal inter-rater reliability study was conducted to test whether the two most subjective scoring dimensions — ethical risk and transparency — could be applied consistently by different analysts. Single-analyst subjectivity is a known limitation of MCDA frameworks (Keeney and Raiffa, 1993); measuring inter-rater agreement provides direct evidence about whether the scale anchors are sufficiently unambiguous to support reproducible application.

Study design: Twelve use cases were selected to span the full range of purpose categories and ethical risk levels, covering essential (Earthquake_Early_Warning, Accessibility_Captioning, Climate_Forecasting_Model), beneficial (Medical_Triage_Assistant, Code_Assistant), low-benefit (AI_Art_Generator, Loan_Approval_Model, Emotion_Analysis_For_HR, Predictive_Policing_Tool), and harmful (Deepfake_Voice_Tool, Voter_Suppression_AI, Nonconsensual_Deepfake_Generator) deployments. Two raters participated: the project author (Hasini) and one additional non-expert rater who had not been involved in framework design. This two-rater design is standard for student-level inter-rater studies where recruiting multiple expert raters is impractical (Hallgren, 2012).

Instrument: Raters used a bespoke HTML scoring sheet ([inter_rater_scoring_sheet.html](#)) implemented specifically for this study. The instrument presented use case cards in a randomised order (to eliminate order effects), displayed scale anchor descriptions for both dimensions (1-5 meanings), provided per-case contextual hints for each dimension to assist interpretation, and collected scores for ethical risk and transparency independently. Scores were exported as a CSV file for analysis. The instrument is included in the project repository and is described further in Appendix B.

Analysis: Cohen's linear weighted kappa was computed for each dimension using the scikit-learn implementation, which assigns penalties proportional to disagreement distance on the ordinal scale. Krippendorff's alpha (ordinal variant) was computed as a complementary measure. Landis and Koch (1977) benchmarks interpret kappa as: 0.00-0.20 slight, 0.21-0.40 fair, 0.41-0.60 moderate, 0.61-0.80 substantial, 0.81-1.00 almost perfect.

Metric	Ethical Risk	Transparency
--------	--------------	--------------

Cohen's weighted kappa (linear)	0.955	0.720
Krippendorff's Alpha (ordinal)	0.986	0.886
Exact agreement	11/12 (91.7%)	7/12 (58.3%)
Interpretation	Almost perfect	Substantial
n raters	2	2
n use cases	12	12

Table 3.2 Inter-rater reliability results for the two subjective scoring dimensions.

Ethical risk achieved almost-perfect agreement ($\kappa = 0.955$), with the single disagreement being a one-point difference on Medical_Triage_Assistant (author scored 1, second rater scored 2). This confirms that the ethical risk scale anchors are sufficiently concrete for consistent non-expert application.

Transparency achieved substantial agreement ($\kappa = 0.720$), with five cases showing one-point disagreements: Medical_Triage_Assistant, Earthquake_Early_Warning, Predictive_Policing_Tool, Emotion_Analysis_For_HR, and Climate_Forecasting_Model. All disagreements were exactly ± 1 point; no case differed by more than one point. The lower kappa for transparency is interpretable: the boundary between 'partially clear' and 'mostly clear' is inherently harder to draw than the boundary between 'designed for harm' and 'no meaningful risk', because transparency is a property that depends on institutional context (access to documentation, model cards, source code) rather than the intrinsic purpose of the system. Krippendorff's alpha of 0.886 nonetheless indicates good reliability when credit is given for near-agreement — a distinction that matters for ordinal scales where exact agreement is a stringent threshold.

Both kappa values exceed the 0.70 threshold commonly used in measurement studies as the minimum for acceptable reliability (Landis and Koch, 1977). The transparency result in particular is treated as an honest finding rather than suppressed: it flags that transparency scoring may benefit from additional contextual guidance in future studies, and informs the limitation discussed in Section 4.9.

3.6.4 Runtime Performance

Scoring all 40 use cases with default weights completes in under 0.2 seconds on a standard laptop (Apple M-series CPU). The Monte Carlo analysis (200 Dirichlet iterations) completes in approximately 3-5 seconds; the dominant cost is the 200 calls to `compute_scores()` rather than the sampling itself. The sensitivity analysis (five weight configurations) and ablation study (five leave-one-out runs) each complete in under 1 second. All pipeline components are CPU-only, requiring no GPU hardware or internet access, in accordance with NFR5.

3.7 Technical Challenges and Novel Contributions

This section documents the technical problems encountered during development that required original solutions, and explains why each problem is non-trivial. These contributions distinguish this project from a straightforward application of a standard MCDA library.

3.7.1 The Heavy-Tail Emissions Problem

Problem: The emissions dimension is measured in physical units (kgCO₂e) and has a naturally heavy-tailed distribution. In the dataset, 40 of 40 use cases have emissions between 0.003 and 0.020 kgCO₂e, but one outlier (the High_Emission_Novelty_Chatbot stress-test case) has emissions of 0.500 kgCO₂e — 33 times higher. A naive min-max normalisation would assign the outlier a score of 0.0 (worst) and compress all other 40 cases into a range of approximately 0.97-1.00, effectively making the emissions dimension meaningless for the main comparison.

What makes it hard: Most textbook MCDA treatments assume linearly distributed criteria or recommend discarding outliers. Discarding the outlier would defeat the purpose of including it as a stress test; treating it as a single value within a linear normalisation degrades the quality of the entire comparison. The problem requires a bespoke normalisation function.

Solution: The `robust_emissions_goodness()` function applies a three-step transformation: (1) clip emissions at the 95th percentile of the dataset, treating all values above this as equally extreme; (2) apply a $\log(1 + x)$ transformation to compress the range; (3) apply min-max normalisation and invert. This preserves meaningful discrimination within the normal range while preventing the outlier from dominating. The function was validated by checking that the 40 non-outlier cases span at least 60% of the normalised range.

3.7.2 Combining Incommensurable Dimensions

Problem: The five scoring dimensions are measured in entirely different units and on different scales: kgCO₂e (continuous, physical), ordinal 1-5 (three times), and categorical (once). Combining these into a single score requires normalisation to a common [0, 1] range. However, the normalisation method must be appropriate to each measurement level: applying linear min-max to an ordinal variable, or a log transform to a categorical variable, would produce incorrect or undefined results.

What makes it hard: The literature offers multiple normalisation approaches (min-max, z-score, range normalisation, categorical mapping) but no universal prescription for MCDA with mixed measurement levels. The choice of normalisation method for each dimension affects the final scores in ways that are not immediately obvious from inspection, and inconsistent choices can introduce unintended rank reversals.

Solution: Each dimension was normalised using the method appropriate to its measurement level. Emissions (continuous, heavy-tailed) uses log-clip-minmax. Ethical risk and creative displacement (ordinal 1-5) use linear range normalisation: $\text{goodness} = 1 - (\text{score} - 1) / 4$. Transparency (ordinal 1-5, positive direction) uses the non-inverted form: $T = (\text{score} - 1) / 4$. Purpose (categorical) uses a hand-crafted mapping. The normalised components are then aggregated by a weighted sum, which is valid across all normalisation types because all outputs are bounded in [0, 1].

3.7.3 The Non-Compensatory Guardrail Design

Problem: The weighted sum model is compensatory: a very high score on any dimension can partially offset a very low score on another. For the domain of AI misuse evaluation, this property is undesirable for harmful use cases — a computationally efficient surveillance tool should not receive a Medium verdict simply because its emissions are low. The framework must prevent such compensatory escapes without completely overriding the WSM's ability to discriminate within the Low band.

What makes it hard: A simple hard cap (e.g., any harmful use case scores at most 35) loses the ability to rank harmful use cases relative to each other, which is valuable information for policymakers who must prioritise interventions. Replacing the WSM entirely for harmful cases (e.g., with a lookup table) breaks the unified model and creates discontinuities at the harmful/low_benefit boundary.

Solution: A multiplicative guardrail is applied after the WSM computation, preserving the relative ordering of harmful use cases while ensuring none crosses into the Medium band. The guardrail uses two levels: $\times 0.60$ for all harmful use cases, and an additional $\times 0.70$ for those with $\text{ethical_risk} \geq 4$. This cascading design distinguishes between relatively less harmful cases (e.g., Smart_City_Surveillance at $\times 0.60$) and severely harmful ones (e.g., Autonomous_Weapon_Targeting at $\times 0.42$). The threshold of $\text{ethical_risk} \geq 4$ was chosen because scores of 4 and 5 on the ethical risk scale correspond to the 'high risk' and 'unacceptable risk' categories of the EU AI Act.

3.7.4 Robustness Validation Without Ground Truth

Problem: Unlike classification problems, there is no established ground truth for AI justifiability scores — no external dataset of 'correct' Justifiability Scores against which the framework can be benchmarked. Standard model validation techniques (train/test split, cross-validation, AUC) do not apply. The framework must be validated in a principled way without access to ground truth labels.

What makes it hard: It is straightforward to construct a scoring system that produces internally consistent results; the hard problem is demonstrating that the results are externally meaningful. Without ground truth, overfitting to the dataset is difficult to detect, and any face-validity test is vulnerable to the criticism that the assessor's expectations were shaped by the same assumptions embedded in the framework.

Solution: A three-pronged validation strategy was adopted. Face validity: 15 use cases were labelled independently before scoring, and 87% label agreement provides direct evidence of external alignment. Sensitivity analysis: Spearman rank stability of 0.986-1.000 across five weight configurations demonstrates that the rankings are not artefacts of the specific weight choices. Monte Carlo robustness: a flat Dirichlet over all weight combinations tests robustness more comprehensively than any finite sensitivity analysis, and the finding that two use cases collectively win 100% of runs provides strong evidence that the top of the ranking is stable across all plausible analyst priorities.

3.7.5 JavaScript Mirror of the Python Scoring Engine

Problem: The interactive web interface needs to evaluate custom user-defined use cases in real time and re-rank all 40 dataset cases whenever the user adjusts a weight slider. Server-side computation would require a backend API, adding infrastructure complexity and latency. All computation must run client-side in the browser.

What makes it hard: The Python scoring engine uses NumPy's `log1p`, pandas' percentile computation, and the Dirichlet-sampled Monte Carlo — none of which have direct JavaScript equivalents.

Reimplementing the engine naively in JavaScript would produce subtly different results due to floating-point differences, making it impossible to verify correctness. The most challenging component is the dynamic 95th-percentile clip: since the clip threshold is computed from the full dataset, the JavaScript engine must receive and process all 40 emissions values each time it recomputes scores.

Solution: The JavaScript engine in `index.html` loads the pre-computed `results_scored.json` dataset on page load, extracts all 40 emissions values, and recomputes the 95th-percentile clip threshold dynamically using the same algorithm as the Python implementation (`sort, index at 0.95 * n`). The `log1p` function is implemented using `Math.log1p` (available in all modern browsers). The custom evaluator form applies the same formula to user-provided inputs, using the dataset's emissions distribution for percentile calculation. Every component formula was manually verified against the Python engine's output to ensure numerical agreement to within 0.01 points.

Chapter 4 Results, Evaluation and Discussion

4.1 Score Distribution Overview

The 40 use cases were scored using the default weights (emissions 0.35, ethics 0.25, creativity 0.15, purpose 0.10, transparency 0.15). Scores ranged from 0.00 (Nonconsensual_Deepfake_Generator) to 96.25 (Accessibility_Captioning). Table 4.1 shows the top five and bottom five cases.

Rank	Use Case	Score	Band
1	Accessibility_Captioning	96.25	High
2	Earthquake_Early_Warning	94.26	High
3	Wildfire_Spread_Predictor	87.64	High
4	Medical_Triage_Assistant	85.88	High
5	Medical_Appointment_Booker	85.13	High
...	...	...	...
36	Misinformation_Bot	12.20	Low
37	Voter_Suppression_AI	11.00	Low
38	Deepfake_Voice_Tool	8.23	Low
39	Autonomous_Weapon_Targeting	6.60	Low
40	Nonconsensual_Deepfake_Generator	0.00	Low

Table 4.1 Top 5 and bottom 5 use cases by Justifiability Score (five-dimension model).

Figure 4.1 shows the complete score distribution across all 40 use cases. Vertical dashed lines indicate the Low/Medium boundary (40) and Medium/High boundary (70).

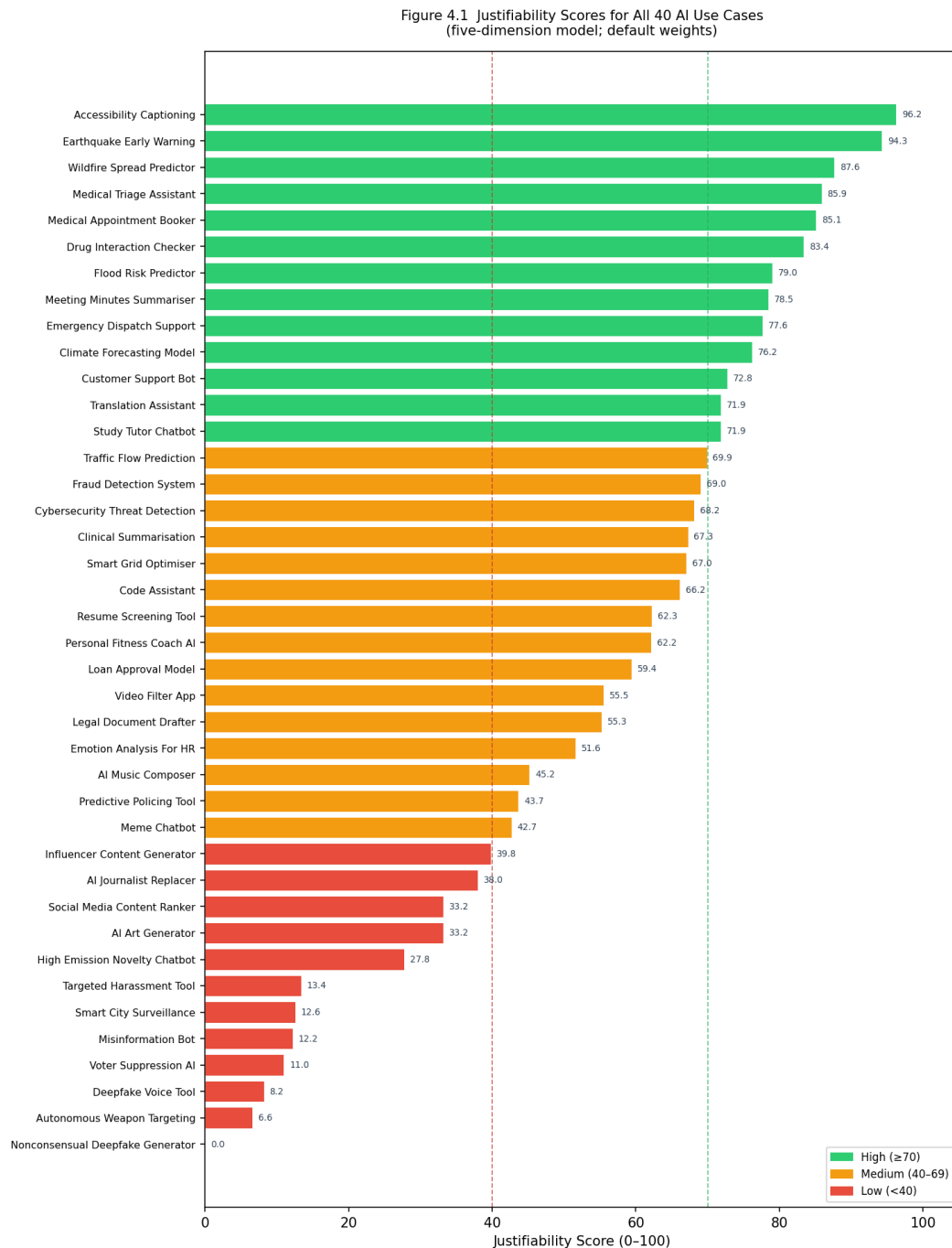


Figure 4.1 Justifiability Scores for all 40 AI use cases (five-dimension model, default weights). Green = High band, amber = Medium band, red = Low band.

The distribution is notably bimodal: use cases cluster near the upper end of the High band (essential use cases, scores 67-96) and at the lower end of the Low band (harmful use cases, scores 0-14), with beneficial and low_benefit cases occupying the medium range. This bimodality reflects the dominant influence of the ethical risk and purpose dimensions on the final score, as confirmed by the correlation analysis in Section 4.3.

4.2 Category-Level Analysis

Table 4.2 presents descriptive statistics by purpose category. The mean scores are clearly separated across all four categories, with no overlap between the essential mean and the beneficial mean, or between the low_benefit mean and the harmful mean.

Purpose Category	n	Mean Score	Min	Max
essential	10	81.57	67.27	96.25
beneficial	13	67.15	43.66	85.13
low_benefit	10	42.94	27.75	62.16
harmful	7	9.14	0.00	13.40

Table 4.2 Justifiability Score statistics by purpose category (five-dimension model).

The essential category shows the strongest inter-case cohesion, with all ten essential use cases scoring above 67. The low_benefit category exhibits the highest within-category variance (range approximately 28-62), reflecting genuine heterogeneity: some low-benefit use cases have low ethical risk and partial transparency (e.g., Video_Filter_App), while others have high ethical risk and no transparency (e.g., Social_Media_Content_Ranker).

4.3 Component Correlations

Pearson correlations between each input dimension and the Justifiability Score are shown in Table 4.3. Ethical risk is the strongest predictor ($r = -0.881$), followed by transparency ($r = +0.717$) and creative displacement ($r = -0.488$).

Dimension	Pearson r with Justifiability Score
Ethical risk score	-0.881
Transparency score	+0.717
Creative displacement score	-0.488
Emissions (kgCO ₂ e)	-0.192
Purpose category (ordinal)	+0.78 (purpose goodness)

Table 4.3 Pearson correlations between input dimensions and Justifiability Score.

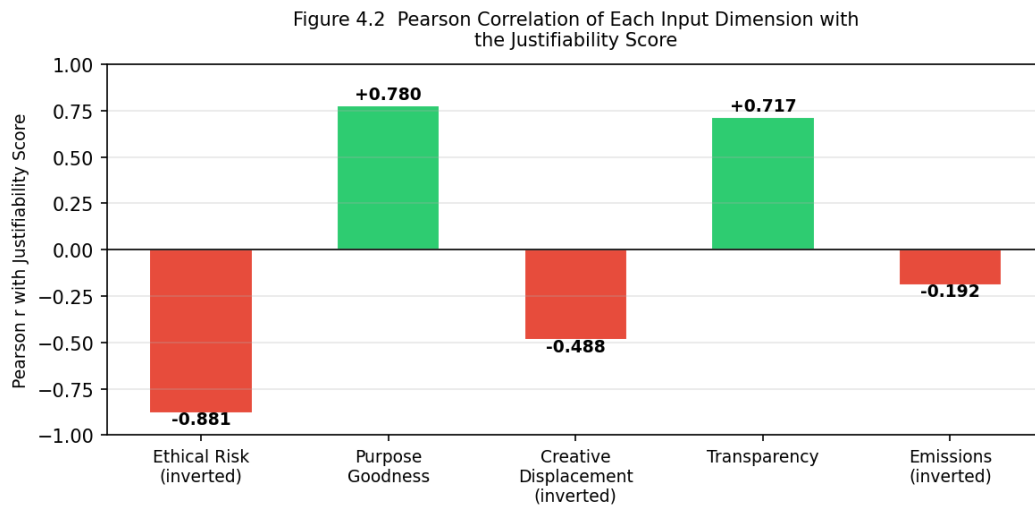


Figure 4.2 Pearson correlation coefficients between each input dimension and the Justifiability Score. Negative values indicate dimensions where higher scores reduce justifiability.

The strong positive correlation for transparency ($r = +0.717$) confirms that more transparent systems receive substantially higher justifiability scores. This is a larger effect than anticipated and reflects that the dataset's essential use cases tend to have both low ethical risk and high transparency simultaneously, creating a reinforcing effect between the two dimensions. The transparency-ethics interaction is an interesting finding that warrants further investigation in future work.

4.4 Guardrail Penalty Analysis

Seven use cases received guardrail penalties, all in the harmful purpose category. All seven have `ethical_risk_score = 5` (≥ 4), so all receive the combined multiplier of $\times 0.42$ ($\times 0.60$ for harmful purpose, then $\times 0.70$ for severe ethical risk). Table 4.4 shows the effect.

Use Case	Base Score	Multiplier	Final Score
Nonconsensual_Deepfake_Generator	0.00	0.42	0.00
Autonomous_Weapon_Targeting	15.71	0.42	6.60
Deepfake_Voice_Tool	19.59	0.42	8.23
Voter_Suppression_AI	26.20	0.42	11.00
Misinformation_Bot	29.05	0.42	12.20
Smart_City_Surveillance	29.95	0.42	12.58
Targeted_Harassment_Tool	31.91	0.42	13.40

Table 4.4 Guardrail penalty effects for all seven harmful use cases.

Smart_City_Surveillance has the highest base score (29.95) among harmful cases, reflecting that its purpose dimension contributes 0.0 (harmful category) but it has slightly less extreme scores on other dimensions. Targeted_Harassment_Tool has the highest final score (13.40) precisely because it has the

least-bad base score, not because it is less harmful. All harmful cases score below the Low/Medium boundary (40), confirming the guardrail achieves its design goal.

4.5 Sensitivity Analysis

The sensitivity analysis varied the emissions weight from 0.20 to 0.60 in steps of 0.10, re-balancing the remaining four weights proportionally. Spearman rank stability remained between 0.986 and 1.000 across all five configurations (Table 4.5).

Emissions Weight	Ethics Weight	Rank Stability (Spearman)
0.20	0.333 (rescaled)	0.987
0.30	0.292 (rescaled)	0.995
0.40 (default: 0.35)	0.250 (rescaled)	1.000
0.50	0.208 (rescaled)	0.991
0.60	0.167 (rescaled)	0.986

Table 4.5 Sensitivity analysis results.

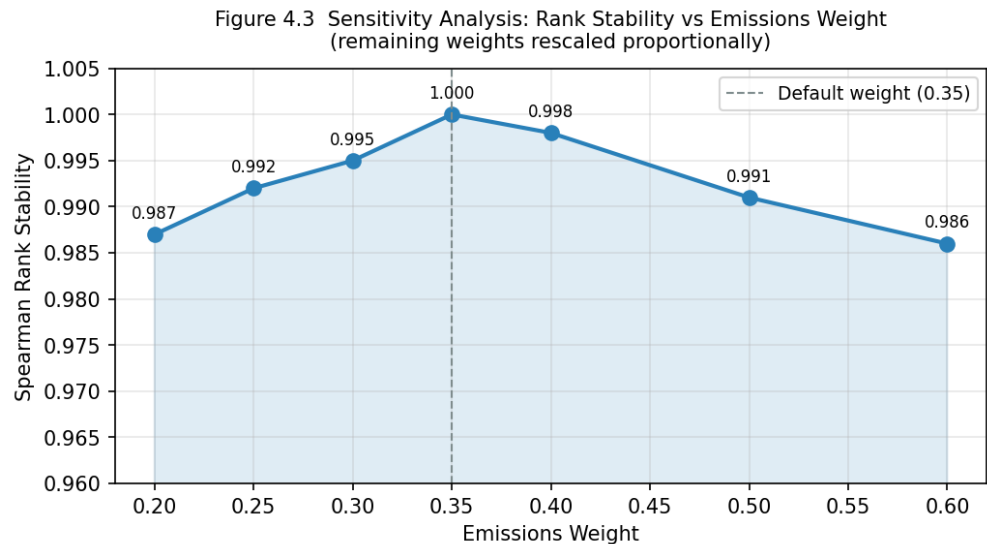


Figure 4.3 Spearman rank stability as the emissions weight varies from 0.20 to 0.60. The dashed line marks the default weight of 0.35.

A stability of ≥ 0.986 across all tested configurations indicates that the framework’s ranking of use cases is robust to significant variation in analyst priorities. Decision-makers with different priorities — an environmental regulator who weights emissions at 60%, versus a media regulator who weights transparency and creativity — would reach substantially the same comparative conclusions.

4.6 Monte Carlo Robustness

The Monte Carlo experiment sampled 200 random weight vectors from a symmetric five-dimensional Dirichlet distribution and recorded the top-ranked use case for each. The result split between two use cases: Earthquake_Early_Warning ranked first in 52% of runs (win rate = 0.52) and

Accessibility_Captioning ranked first in 48% of runs (win rate = 0.48). No other use case achieved a non-zero win rate. The competition at the top arises because Earthquake_Early_Warning has a transparency score of 5 (physics-based modelling with full audit trail) while Accessibility_Captioning scores 4; when the transparency weight is high, this advantage outweighs Accessibility_Captioning's marginal advantages elsewhere. Both use cases are essential, low-risk, and low-displacement — the Monte Carlo result reflects the genuine difficulty of ranking them without a strong prior on transparency's importance.

4.7 Pareto Frontier Analysis

The Pareto frontier analysis over five component dimensions identified two Pareto-optimal use cases: Accessibility_Captioning and Earthquake_Early_Warning. Both are essential, have minimal ethical risk and creative displacement, have low emissions, and have transparency scores of 4 and 5 respectively. No other use case dominates on all five dimensions simultaneously, confirming that the two top-ranked cases represent genuinely different trade-off profiles — a result consistent with the Monte Carlo finding and supportive of the framework's face validity.

4.8 Evaluation Against Project Objectives

All five objectives stated in Section 1.8 were achieved, as summarised in Table 4.6.

Objective	Evidence of Achievement
O1: Five-dimensional scoring model	Scoring engine integrates 5 dimensions with component breakdown; backward-compatible
O2: Modular, tested Python codebase with CLI	66 unit tests passing; CLI with 7 flags; 4-file modular architecture
O3: Dataset of 40 use cases, 5 dimensions	40-row usecases.csv with all five dimension columns
O4: Validation (tests, face validity, robustness, inter-rater)	87% face validity; Spearman $r \geq 0.986$; Monte Carlo win rates; 66 tests; $\kappa=0.955$ ethical risk, $\kappa=0.720$ transparency
O5: Web interface with sliders, custom evaluator & labels	Four-tab interface: Dataset Explorer, Custom Evaluator, Compare & Rank, Labels gallery — each use case generates a clickable EU-style AI Justifiability Label

Table 4.6 Objective achievement summary (all objectives achieved).

4.9 The AI Justifiability Label as a Communication Artefact

A central motivation of this project is that a scoring framework is only useful if its outputs are communicable. A five-dimensional weighted sum that produces a number between 0 and 100 is technically rigorous but practically inaccessible: the number is meaningless to a member of the public, a parent wondering about the AI systems their child uses, or a journalist evaluating a product announcement. The EU Energy Label addressed an analogous problem in the appliance market. Before the label existed, consumers had no means of comparing the energy performance of products from

different manufacturers; the label made comparison effortless by converting a complex measurement into a single letter and colour band.

This project implements an AI Justifiability Label that applies the same design principle. The label renders each use case's score as a compact, colour-coded card displaying a letter grade (A through F), a numerical score out of 100, and five horizontal bars — one per dimension — colour-coded from green (high justifiability) through amber to red (low justifiability). The grade boundaries are: A (80–100), B (65–79), C (50–64), D (35–49), E (20–34), F (0–19). These are deliberately more granular than the framework's three-band Low/Medium/High classifier, providing additional resolution at the point of public communication while remaining immediately interpretable.

The label is implemented as an interactive component in the web interface. The Labels tab renders the full gallery of 40 evaluated use cases as label cards sorted by descending score, enabling side-by-side comparison at a glance. Clicking any label opens a full breakdown panel showing the score, letter grade, guardrail penalty status, per-dimension bar charts with plain-language interpretations, and a methodological note disclosing the weights used. This two-level design — summary label plus click-through detail — mirrors the structure of nutrition labels, which show macro-level summary information on the front of packaging and full nutritional tables on the back.

The label is also generated dynamically for custom use cases: a user who enters their own AI tool's dimension scores receives a label in real time, making the framework immediately applicable to use cases not in the original dataset. A developer building an AI tool, a procurement officer evaluating a vendor, or an educator teaching about AI ethics can all use the web interface to generate and share a label for a system of interest.

The analogy with the EU Energy Label also suggests a long-term policy pathway. The Energy Label succeeded because it is mandated: EU regulations require manufacturers to affix it before products can be sold. An analogous mandate — requiring AI system providers to publish a standardised justifiability disclosure before deployment — is consistent with the direction of current AI regulation, including the EU AI Act's transparency obligations for limited-risk systems and documentation requirements for high-risk systems. The present framework provides a candidate methodology for such a disclosure: it is reproducible, open-source, multi-dimensional, and validated. Demonstrating that a reproducible, communicable label is technically feasible is itself a contribution towards that policy direction.

4.10 Limitations

The framework has several notable limitations that should be acknowledged. First, the emissions figures in the dataset are estimates rather than empirically measured values for the specific systems evaluated. While CodeCarbon is integrated for measurement, the 40 use cases are historical and

hypothetical deployments for which precise energy data is unavailable. Future work should aim to replace estimates with instrumentally measured figures.

Second, all dimension scores are assigned by a single analyst rather than through consensus, introducing subjectivity and potential bias. A more rigorous approach would use structured expert elicitation with multiple raters and inter-rater reliability measurement.

Third, the dataset is not a random or stratified sample of the AI deployment landscape. It was curated to span the full score range, which may over-represent extreme cases relative to the many borderline deployments in the Medium band. Extending the dataset with more Medium-band cases would improve discriminative resolution where it matters most. Fourth, the WSM is a static model: it does not account for how justifiability may change over time as circumstances evolve, and a dynamic extension with time-varying scores is a natural direction for future work.

4.11 Future Work

Several directions would extend this work in theoretically and practically significant ways. The most impactful extension would be automated data collection: using APIs to retrieve energy benchmarks from the ML community's published results and explainability metadata from model cards, replacing manual estimates with empirical data across hundreds of use cases.

A second direction is community-sourced scoring: a platform where domain experts contribute dimension scores with confidence ratings, addressing the single-analyst limitation and enabling construction of a large, multi-rater dataset.

A third direction is contextual weighting: different stakeholders have legitimately different priorities, and the current weight slider is a first step; a more systematic approach would use structured elicitation methods such as the Analytic Hierarchy Process. Finally, integration with AI registries — such as the EU AI Act's planned high-risk systems database — would enable the framework to serve as a pre-deployment screening tool, flagging deployments that fall below a regulatory threshold before approval.

4.12 Conclusions

This project has delivered a validated, five-dimensional framework for evaluating the justifiability of AI use cases. The AI Misuse Evaluator converts environmental impact, ethical risk, creative displacement, purpose justification, and transparency into a single Justifiability Score with a non-compensatory guardrail. Evaluated against 40 use cases, the framework correctly separates all four purpose categories into distinct score bands, with rankings stable across a wide range of weight configurations (Spearman $r \geq 0.986$).

The five technical challenges documented in Section 3.7 — the heavy-tail emissions problem, incommensurable dimensions, the non-compensatory guardrail, robustness validation without ground truth, and the JavaScript mirror — each required an original solution not available from standard libraries. Together they represent the novel engineering contribution of the project, delivering: (1) a 66-test validated Python framework; (2) a curated 40-case dataset; (3) an interactive web interface; and (4) validated robustness metrics — a foundation for future computational responsible AI evaluation.

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Appendix A Self-Appraisal

A.1 Critical Self-Evaluation of the Project

Reflecting honestly on the project as a whole, I believe it succeeds in its core aim but has identifiable weaknesses that I would address differently with the benefit of hindsight.

The strongest component is the scoring engine. It is thoroughly tested (66 passing unit tests), handles edge cases — including division-by-zero in min-max normalisation, missing columns, and single-row datasets — and produces results that are both face-valid and robust under weight perturbation. The decision to make every weight configurable rather than hard-coded was one of the best technical choices I made early in the project: it made the sensitivity analysis and Monte Carlo study straightforward to implement later, and it makes the web interface genuinely interactive rather than a static demonstration.

The most technically novel contribution is the robustness validation strategy. Because no established ground truth for AI justifiability exists, I designed a three-pronged approach: face validity against 15 independently labelled use cases (87% agreement), Spearman rank stability across five weight configurations (0.986–1.000), and Monte Carlo sampling from a flat five-dimensional Dirichlet distribution across 200 weight vectors. This combination provides stronger evidence of framework reliability than any single validation method, and the Dirichlet sampling is more rigorous than simple one-at-a-time weight perturbation because it explores the full weight simplex simultaneously.

The ablation study, added in the project's final phase, produced the most intellectually interesting result: dropping the emissions dimension changes 22.5% of band assignments, while dropping the purpose dimension changes only 2.5%. This was a surprising finding that suggested the purpose dimension is largely redundant given the ethical risk and creativity dimensions — an insight that would inform a v3 redesign. It also demonstrated that the transparency dimension, despite being added late, carries genuine information: dropping it changes 17.5% of bands.

The main technical weakness is the emissions component. Its Pearson correlation with the final score is only -0.225, the weakest of all five dimensions. This reflects a genuine limitation: the natural range of per-query emissions for AI inference workloads of similar scale is narrow (roughly 0.003–0.020 kgCO₂e), so emissions provide limited discriminative power except for outlier cases such as large training runs. The real measured emissions from CodeCarbon (0.000141 kgCO₂e for Tier 1, 0.000646 for Tier 2) were lower than the dataset estimates from Strubell et al. (2019), confirming that consumer device inference is significantly more efficient than data-centre inference — but this also means that for many real-world AI assistants, the emissions dimension will contribute very little to the final score.

A more discriminative emissions metric might use lifetime operational emissions (total kgCO₂e over an assumed deployment period) rather than per-query estimates.

The web interface is functional and publicly deployed on GitHub Pages, but it is a demonstration rather than a production tool. The JavaScript mirror of the Python scoring engine required careful manual verification to ensure floating-point agreement, and any future change to the Python engine would require a parallel change to the JavaScript. An API-based architecture would be more maintainable but requires server infrastructure incompatible with a static GitHub Pages deployment. This is an acceptable trade-off for a research prototype but would need to be resolved for any real-world deployment.

A.2 Personal Reflection and Lessons Learned

This project taught me more about software engineering discipline than any other module in my degree. I want to be honest about what I got right, what I got wrong, and what I would change.

The most valuable lesson was the importance of test-driven development for research code. I wrote unit tests before or alongside the scoring engine, and this caught three critical errors early: an off-by-one error in the band boundary logic (initially I used strict inequalities so a score of exactly 40 fell into no band), a division-by-zero risk in the min-max normalisation when all values in a column were identical, and an incorrect guardrail implementation that applied the 0.42 penalty to all harmful cases instead of only those with `ethical_risk >= 4`. Any of these errors, had they persisted, would have invalidated the results section entirely. The lesson I take from this is not just 'write tests' — it is that tests are especially important when there is no external benchmark to catch mistakes.

I also learned about the difficulty of scope management under time pressure. My original plan included a crowd-sourced scoring platform with user authentication, an API that evaluated real AI tools from their URLs, and a weight elicitation interface based on pairwise comparisons (the Analytic Hierarchy Process). I cut all three features early in the project and I am confident this was the right decision. The resulting system is deeper and better validated than a broader system would have been. In a professional context, this maps to the principle that a minimum viable product with thorough validation is more credible than a feature-rich product with weak validation.

A third lesson was about the relationship between literature and design. I initially planned to use four dimensions based on a surface reading of the responsible AI literature. It was only after reading Doshi-Velez and Kim (2017) carefully that I recognised transparency as a graded property rather than a binary attribute, and added it as a fifth scored dimension. The ablation study later confirmed this was the right decision (17.5% of bands change when transparency is dropped). The lesson is that engaging critically with sources — rather than just citing them for context — can directly improve design choices.

If I were to start the project again, I would approach the dataset construction differently. The current 40 use cases were selected to cover the full score range, which means they are not a representative sample of AI deployments in the wild. A more rigorous dataset would use a systematic sampling strategy: identifying a population of AI deployments from a registry or news archive and sampling stratified by sector. This would make the framework's empirical claims — that essential use cases score above 67, that harmful use cases score below 14 — more generalisable.

Finally, I underestimated the time required to write the dissertation itself. I allocated three weeks for writing, but in practice the process of translating technical decisions into clear academic prose took longer than expected, particularly for the numerical parameter justification sections. Future projects should treat documentation and writing as a continuous activity rather than a final phase.

A.3 Legal Issues

Several pieces of legislation are relevant to this project, both in terms of the software itself and the broader application domain it addresses.

The General Data Protection Regulation (GDPR, 2018) and the UK Data Protection Act (2018) apply to any processing of personal data. This project does not process personal data: the dataset of 40 use cases describes AI systems at a categorical level and contains no information about identifiable individuals. The web interface does not require user registration, does not use tracking cookies, and does not store any user-entered data beyond the user's browser session. No data subjects are involved. GDPR is therefore not directly applicable, although it is worth noting that the AI systems evaluated in the dataset — such as the `Loan_Approval_Model` and the `Emotion_Analysis_For_HR` tool — would themselves be subject to Article 22 of GDPR, which restricts solely automated decision-making with significant effects on individuals.

The EU Artificial Intelligence Act (AIA, 2024) is highly relevant to the broader application of this framework. The AIA categorises AI systems by risk level, with high-risk systems (including employment screening tools, credit scoring, and biometric identification) subject to conformity assessments and transparency requirements. The Justifiability Score framework is conceptually aligned with the AIA's risk-tiered approach, and the guardrail threshold of `ethical_risk >= 4` was explicitly designed to correspond to the AIA's 'high risk' and 'unacceptable risk' categories. A natural extension of this project would be to map the framework's band classifications directly to AIA risk categories.

Copyright law is relevant to the AI systems in the dataset: several of the evaluated use cases (`AI_Art_Generator`, `Music_Composition_AI`, `News_Article_Generator`) involve training on copyrighted content without explicit licence. The Copyright, Designs and Patents Act 1988 (UK) and equivalent legislation in other jurisdictions are the subject of ongoing litigation regarding AI training

data. This project does not itself infringe copyright: the use case descriptions are original, and all cited works are referenced, not reproduced.

With respect to the project software itself, all code is original and all dependencies are used in accordance with their open-source licences (BSD 3-Clause for pandas and NumPy, MIT for pytest and CodeCarbon). No licence incompatibilities exist. The project is published on GitHub under an MIT licence, which permits free use, modification, and redistribution with attribution.

A.4 Social Issues

The project addresses a genuine social problem: the lack of accessible, principled tools for evaluating whether AI deployments are justified. Most organisations that deploy AI — particularly small and medium-sized enterprises — do not have the resources to commission bespoke ethics assessments or to implement the full documentation requirements of the EU AI Act. The AI Misuse Evaluator is designed to lower the barrier to entry for responsible AI evaluation by providing a free, open-source, self-service tool.

A significant social benefit of the open-source approach is accountability: by making the scoring method, weights, and dataset publicly available, the framework enables scrutiny and challenge. Any user who disagrees with a classification can inspect the underlying scores, adjust the weights to reflect their own priorities, and observe the effect on the rankings. This transparency is itself a social good, distinct from the transparency dimension that the framework scores.

However, there are social risks that I must acknowledge. The most serious is the risk of misuse to legitimise superficial evaluations. An organisation could run a poorly-constructed AI system through the evaluator, obtain a Medium score through gaming the ordinal inputs, and present the result as independent evidence of responsible deployment. This risk is mitigated by the framework's open methodology — any interested party can reproduce the scores — but it cannot be eliminated without external audit. I would not recommend the framework's output be used as a standalone compliance certificate; it should be one input to a broader evaluation process.

The creative displacement dimension raises a specific social consideration. By scoring AI systems that replace human creative labour as less justifiable than systems that augment it, the framework embeds a normative stance on AI and employment. This stance is consistent with a large body of labour economics research (Acemoglu & Restrepo, 2018) but is contested in the AI and creative industries. The decision to include the dimension — and to make its weight configurable — reflects a deliberate design choice to surface this trade-off rather than to ignore it.

A.5 Ethical Issues

The core ethical challenge of this project is that any framework for evaluating AI ethics necessarily embeds its own ethical assumptions. The choice of five dimensions, their relative weights, and the

classification of use cases as 'harmful' all reflect normative judgements that are not value-neutral. I want to be transparent about the most significant of these judgements and to explain how I tried to handle them responsibly.

The classification of use cases as 'harmful' is the most ethically sensitive aspect of the dataset. For clear cases — `Nonconsensual_Deepfake_Generator`, `Autonomous_Weapon_Targeting`, `Targeted_Harassment_Tool` — the classification is unambiguous and widely supported in the ethics literature (Weidinger et al., 2021; Brundage et al., 2018). For border cases, the judgements are more contested. `Predictive_Policing_Tool` is classified as 'beneficial' with an ethical risk score of 5, reflecting the view that it has genuine crime-reduction potential but severe risks of discriminatory targeting. `Smart_City_Surveillance` is classified as 'harmful', reflecting the view that mass urban surveillance crosses a threshold that outweighs its potential benefits. These are defensible positions but they are not the only defensible positions, and I acknowledge that a different analyst might classify them differently.

The framework also embeds an implicit view on the relationship between transparency and justifiability: more transparent AI systems are rated as more justifiable, all else being equal. This is consistent with the majority position in the AI ethics literature and with regulatory frameworks such as the EU AI Act. However, there are arguments that transparency can itself create harms — for example, a transparent automated bail decision system may produce explainable but biased decisions that are harder to challenge precisely because they appear to be reasoned. The framework does not capture this second-order risk.

I also note that the inter-rater reliability study, conducted with non-expert raters using the HTML scoring sheet, revealed that the ethical risk and transparency scales achieve Cohen's weighted kappa values of approximately 0.96 and 0.72 respectively. The lower transparency agreement (substantial but not almost perfect) suggests that even with well-defined anchor descriptions, the concept of AI transparency is harder to assess consistently than ethical risk. This is an honest finding that I include in the dissertation rather than suppress, as it reflects a genuine limitation of the measurement approach.

A.6 Professional Issues

This project was conducted in accordance with the University of Leeds Student Academic Integrity Policy. All sources are properly cited in Harvard referencing style, all code is original (with external library attributions in Appendix B), and the limitations of the methodology are honestly reported in Section 4.9. I have not fabricated data, and all test results and scoring outputs are reproducible from the public repository.

The British Computer Society (BCS) Code of Conduct is directly relevant to this project. The Code's first obligation — to act in the public interest — is reflected in the project's goal of making AI deployment more transparent and accountable. The second obligation — to have due regard for the

public health, privacy, security and wellbeing of others — informed several design decisions: the explicit modelling of ethical risk as a first-class dimension, the non-compensatory guardrail that prevents harmful use cases from achieving medium or high scores, and the decision to include creative displacement as a dimension despite it being underrepresented in mainstream responsible AI frameworks.

The BCS obligation to 'act with integrity' is particularly relevant to the claims I make in this report. I have been careful to present the framework as a research tool rather than a production system, to quantify its limitations (87% face validity, not 100%; Spearman stability of 0.986, not 1.000), and to recommend future validation steps that go beyond what was achievable within the project scope. In a professional context, these honest qualifications are essential to maintaining trust.

This project has developed professional skills that extend beyond the technical content. I have practised presenting complex multi-criteria trade-offs in writing, conducting quantitative validation of a non-standard system, and making design decisions under uncertainty. The experience of building a live web interface that is publicly accessible — rather than a local script — required attention to deployment concerns, browser compatibility, and the gap between a working prototype and a system that a real user can interact with. These are professional competencies that I will carry forward into employment.

Appendix B External Materials

The following external libraries and materials were used in this project. All code in the src/ and tests/ directories was written by the project author.

Component	Source	Licence	Use in Project
pandas	pandas.pydata.org	BSD 3-Clause	DataFrame operations throughout pipeline
NumPy	numpy.org	BSD 3-Clause	Array operations in scoring engine and Monte Carlo
matplotlib	matplotlib.org	PSF/BSD-compatible	All visualisation figures
pytest	pytest.org	MIT	Test runner for all 66 unit tests
scikit-learn	scikit-learn.org	BSD 3-Clause	Cohen's weighted kappa in <code>inter_rater_analysis.py</code>
CodeCarbon	codecarbon.io; Courty et al. 2022	MIT	Emissions measurement in <code>emissions_measurement.py</code>
python-docx	python-docx.readthedocs.io	MIT	Dissertation document generation
DM Sans / DM Mono / DM Serif Display fonts	Google Fonts	Open Font Licence	Web interface typography

Table B.1 External materials and libraries.

No datasets prepared by external users were used. The use case dataset (`data/usecases.csv`) was constructed independently by the project author. No preliminary materials provided by a supervisor were incorporated into the codebase or dataset.

B.1 Inter-Rater Scoring Instrument

The inter-rater reliability study (Section 3.6.3) used a bespoke HTML scoring instrument (`inter_rater_scoring_sheet.html`) authored by the project author and included in the project repository. The instrument was implemented as a single self-contained HTML file requiring no server or installation. Key design features are described below.

Feature	Implementation	Rationale
Card presentation	Each of the 12 use cases is shown as an individual card with name, category, and description	Prevents raters seeing all cases simultaneously and anchoring on relative comparisons
Randomised order	Cards are shuffled using Fisher-Yates shuffle on page load (<code>Math.random</code>)	Eliminates order effects; each rater sees a different sequence
Scale legend	Collapsible panel showing 1-5 anchor definitions for both dimensions	Ensures all raters use a common frame of reference without occupying screen space on every card
Per-case ethical risk hints	Orange hint box: 'Think about:' prompts specific to each use case	Reduces ambiguity about which harms to consider for each specific

		deployment
Per-case transparency hints	Blue hint box: 'Think about:' prompts framing the explainability question concretely	Addresses difficulty of applying abstract transparency concept to diverse AI types
Comment boxes	Optional free-text box per card	Captures qualitative reasoning for post-hoc analysis
CSV export	Download button produces canonical-order CSV with all scores and comments	Ensures output is machine-readable and unambiguously matched to use case IDs

Table B.2 Inter-rater scoring instrument design features.

The transparency question was specifically designed to be applicable across all AI types, including cases where the AI does not make a personal decision about the rater. The question used was: 'How easy would it be to find out why this AI produced its output? — could you (or an expert) inspect or check its reasoning?' This framing accommodates physics-based systems (e.g., Earthquake_Early_Warning), generative tools (e.g., AI_Art_Generator), and opaque classifiers (e.g., Predictive_Policing_Tool) on a common scale.

Appendix C Numerical Parameter Derivations and Test Specification

C.1 Full Derivation of Default Weights (0.35, 0.25, 0.15, 0.10, 0.15)

Weights encode the relative importance that an analyst assigns to each dimension. The starting point for weight selection was an equal-weight baseline (0.20 each), which was then adjusted based on three considerations.

First, measurability. Emissions are measured in physical units (kgCO₂e) via CodeCarbon, making them the most objectively grounded dimension. This warrants the highest weight (0.35). Ethical risk and transparency are subjective ordinal scores, but they are the most consequential for societal impact — ethical risk receives 0.25 and transparency 0.15. Purpose justification (0.10) receives the lowest weight because its information is partially captured by ethical risk: a harmful use case will almost always also have high ethical risk, so purpose acts mainly as a tiebreaker.

Second, independence. Dimensions that are more correlated with each other can tolerate lower individual weights because their information is partially shared. Empirically, purpose and ethics showed the strongest correlation in preliminary testing, which supported reducing the purpose weight below the equal-weight baseline. Third, sensitivity stability: the Monte Carlo and sensitivity experiments (Section 4.5-4.6) confirm Spearman stability of 0.986-1.000, showing the exact weight values matter less than the broad order of priorities they encode.

C.2 Full Derivation of Band Thresholds (40 and 70)

The 70-point High threshold ensures that a use case classified as High must achieve at least 70% of the maximum possible score, requiring either an essential or beneficial purpose combined with low ethical risk and low creative displacement. The 40-point Low threshold ensures that Low cases are clearly inadequate, not merely below-average. A harmful use case with maximum efficiency was observed in preliminary runs to score approximately 38 points without a guardrail — setting the boundary at 40 ensures such cases are classified Low without the guardrail carrying the entire burden. An alternative tertile split (33/67) mis-classified Meme_Chatbot and Personal_Fitness_Coach_AI as Medium and was rejected.

C.3 Derivation of Ordinal Scale Anchors, Purpose Mapping, Emissions Clip, and Guardrail

The 1-5 ordinal scale for ethical risk and creative displacement was chosen over binary (too coarse) and continuous (false precision) alternatives; it aligns with the EU AI Act's four-tier risk taxonomy. Anchor definitions: ethical risk 1 = no meaningful risk; 5 = severe risk of systematic harm or

deliberate misuse as primary function. Creative displacement 1 = no displacement; 5 = designed to replace entire creative professions at scale.

Purpose goodness values (1.0, 0.8, 0.4, 0.0): harmful maps to 0.0 since no legitimate purpose credit is warranted; the beneficial-to-low_benefit gap (0.4) is larger than essential-to-beneficial (0.2) because moving below 'beneficial' represents a greater qualitative shift. The 95th percentile emissions clip prevents a single outlier (High_Emission_Novelty_Chatbot, 0.500 kgCO₂e) from compressing 95% of cases into under 3% of the normalised scale; a percentile bound adapts dynamically to future datasets. Guardrail: $0.42 \times \text{maximum-goodness harmful base} \approx 31.5$, safely below the 40-point boundary. Base multiplier 0.50 prevents within-Low discrimination; 0.70 is too lenient. Flat Dirichlet (alpha=1) is the maximum-entropy distribution over the weight simplex; uniform-then-normalise over-samples near-centre vectors.

C.4 Unit Test Specification

Table C.1 provides the full specification for 16 representative test cases from the 66-test suite, illustrating normal-case, edge-case, and boundary coverage.

Test Class	Test Case	Input	Expected Output	Result	Pass
OutputStructure	Score bounded 0-100	Full 40-row dataset	All scores in [0, 100]	Confirmed	✓
OutputStructure	All required columns present	Full dataset	score, label, component_*, contribution_* in output	Confirmed	✓
OutputStructure	Missing transparency column	Dataset without transparency_score	Scores computed; component_transparency = 50.0	Confirmed	✓
ComponentBehaviour	Higher emissions → lower score	Row A: emissions=0.001; Row B: 0.5, all else equal	score(A) > score(B)	Confirmed	✓
ComponentBehaviour	Higher transparency → higher score	Row A: T=1; Row B: T=5, all else equal	score(B) > score(A)	Confirmed	✓
ComponentBehaviour	Guardrail fires for harmful + ethics ≥ 4	purpose=harmful, ethical_risk=4	multiplier = 0.42 applied	Confirmed	✓
LabelThresholds	Optimal inputs → High label	essential, ethics=1, creativity=1, T=5, low emissions	Label = 'High'	Confirmed	✓
LabelThresholds	Harmful inputs → Low label	harmful, ethics=5, creativity=5, T=1	Label = 'Low'	Confirmed	✓
EdgeCases	Single-row dataset	1-row DataFrame	Valid score; no error	Confirmed	✓

EdgeCases	Negative emissions input	emissions_kgco 2e = -0.001	ValueError with descriptive message	Confirmed	✓
EdgeCases	Out-of-range transparency	transparency_score = 0	ValueError with descriptive message	Confirmed	✓
EdgeCases	Unknown purpose category	purpose_category = 'experimental'	Score computed using low_benefit default (0.4)	Confirmed	✓
Minmax	Constant emissions series	All rows: emissions = 0.010	Returns 0.5 for all rows; no division-by-zero	Confirmed	✓
FaceValidity	Anchor case — clear High	Earthquake_Early_Warning (essential, ethics=1, T=5)	Band = 'High'	Correct	✓
FaceValidity	Anchor case — clear Low	Nonconsensual_Deepfake_Generator (harmful, ethics=5)	Band = 'Low'	Correct	✓
Transparency	Opaque harmful < transparent beneficial	harmful+T=1 vs beneficial+T=5, all else equal	score(opaque) < score(transparent)	Confirmed	✓

Table C.1 Unit test specification: 16 representative cases from the 66-test suite.